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Source-Entropy-Correlated Recoverability of CNC Toolpath Coordinates under Low-Rate Sensing: Quantization Structure, an Operation-Class Ordering, and the 4 Hz Sub-Nyquist Floor

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Abstract: How much of a CNC toolpath-coordinate stream is recoverable *in principle* from low-rate (4 Hz) multi-modal sensing — the question underlying any sensor-side forensic control — has not been characterized. We establish a **source-entropy-correlated recoverability ordering** anchored by two limits. First, the **Shannon source entropy** of the coordinate corpus, measured *directly on raw CAM G-code* with no model in the loop: high-order digits are near-deterministic while fractional digits plateau near the $\log_2 10 \approx 3.32$-bit uniform ceiling — the source precision floor. A separate *representation* effect: the V8 tokenizer's 0.001 grid discards the X/Y/Z ten-thousandths (R retains it), so that "structural zero" is not a source or sensor limit. The fine-digit ceiling stratifies by operation class at the thousandths (facing 2.10 bits; pocketing/adaptive at the ceiling). Second, a **4 Hz sub-Nyquist floor**: 77.7% of G-code blocks occupy ≤ 1 sample period. We report the marginal $H(X)$ (not a Fano bound) and confirm it on a float-epsilon-corrected 5-fold retrain: across 30 per-(axis, slot) cells recovery accuracy inversely tracks entropy ($r = -0.83$, $n = 30$; pooled six-slot $r = -0.89$, illustrative at $n = 6$). Scope is single-platform, closed-vocabulary, teacher-forced; the companion decoder is one existence proof.

Keywords: source entropy; information theory; CNC machining; G-code; toolpath generation; quantization; Nyquist sampling; manufacturing cybersecurity; recoverability ordering; Miller–Madow correction

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1. Introduction

A modern CNC mill executes a toolpath specified by a G-code program: an ordered stream of motion commands and address–value pairs (`X`, `Y`, `Z` coordinates; `R` arc radii; `F` feedrates) that the controller interpolates into multi-axis motion. Every cut radiates a multi-modal physical signature — vibration, acoustic emission, motor current, ambient conditions — that is in principle informative about the instruction stream that produced it. A natural and consequential question follows: *how much of the executing toolpath can a low-rate sensor stream recover, in principle, regardless of the recovering machinery?* The answer characterizes what any sensor-side forensic audit of a contested CNC run can hope to achieve, and it is a property of the source and the sensing channel, not of a particular neural network.

This paper characterizes a **source-entropy-correlated recoverability ordering** anchored by two genuine limits. The informational anchor is the Shannon source entropy of the toolpath-coordinate corpus itself, resolved per primary axis (X, Y, Z), per decimal digit-slot, and per machining operation class (we report the marginal $H(X)$, not a Fano bound; see below). The two places where a genuine limit holds are (i) the slots whose source entropy is identically 0 bits (a digit the source never varies cannot be recovered wrongly, and equally carries no information to recover), and (ii) the physical floor: a sub-Nyquist sampling limit imposed by a low-rate sensor stack, under which the majority of G-code blocks execute faster than one sample period and cannot be resolved as distinct transients. Both the entropy structure and the Nyquist floor are computed directly from the source — the raw CAM-authored G-code and the measured per-row execution durations — and neither depends on any recoverer. We then *demonstrate* the ordering empirically: the companion decoder of [1] and an independent from-scratch sequence-to-sequence baseline both reach the same accuracy *on the categorical command head*, and per-field accuracy tracks per-field source entropy. The decoder is therefore an **existence proof** that the ordering is attainable, not the object of study: this paper owns the *entropy structure*; the companion paper [1] owns the *decoder*.

Why this is an information-theoretic question, not a decoder question.

Recoverability of a symbol from a noisy observation is bounded below in error probability by the *conditional* entropy of the symbol given the observation (Fano's inequality [2,3]). We are deliberately careful here: we measure the marginal source entropy $H(X)$, not $H(X \mid \text{observation})$, so the source entropy is not by itself a Fano bound on accuracy. What it does deliver are two clean endpoints — a source digit with 0 bits of entropy carries no recoverable information and is trivially "recovered" as the constant it always is, while a digit at the uniform ceiling ($\log_2 10 \approx 3.32$ bits) leaves a recoverer at the modal-class prior unless the channel is informative about it — and, between those endpoints, an empirical *ordering*: per-field accuracy falls monotonically as source entropy rises. The classical instance of source-entropy analysis is Shannon's prediction-of-printed-English experiment [4], which estimates a per-symbol entropy of a structured discrete source; we port that measurement to the CNC instruction stream. The novelty is not the inequality but the *object*: we show that the per-field recoverability of a CNC toolpath is ordered by the empirical source entropy of CAM-authored coordinates, that this entropy has a sharp and physically interpretable structure (integer places deterministic, fine decimals plateauing near the uniform ceiling across the hundredths–thousandths), and that the structure stratifies by toolpath-generation strategy. To our knowledge no prior work quantifies the source entropy of CNC G-code coordinates as a recoverability ordering, in either the subtractive-machining or additive-manufacturing literature; we treat the absence of prior art as documented due diligence rather than asserted novelty (Section 2).

The salami-defense and the apparatus.

A reader may ask whether this is merely a relocated analysis of the companion decoder's predictions. It is not: the central object — the source-entropy structure — is computed *decoder-free, on raw G-code source text*, by a dedicated apparatus (Section 4) that never touches the model, the tokenizer, or the decoder's target encoding. The entropy structure reproduces from raw source to within $\sim$0.07 bits of the model-side estimate at the hundredths–thousandths ceiling (the load-bearing X/Y/Z slots); the coarse integer places and the tokenizer-discarded ten-thousandths differ more by construction (Section 4). The entropy ordering exists whether or not anyone builds a recoverer; the recoverer only confirms that the ordering is attainable.

Contributions.

(1) **A decoder-free source-entropy apparatus** for CNC toolpath coordinates (Section 4): per-(axis, slot), per-(axis, slot, op-class), and per-axis sign entropy computed directly from raw CAM G-code, establishing the entropy structure as a property of the source files.

(2) **A three-regime decomposition of the fine-digit entropy ceiling** (Section 5): facing / pocketing / adaptive-HSM operations occupy distinct, mechanistically explained bands of thousandths-digit entropy, traceable to CAM toolpath-generation strategy.

(3) The 4 Hz sub-Nyquist physical floor (Section 6): a measured per-row execution-duration audit showing 77.7% of blocks occupy ≤ 1 sample period at the released sensor rate — one of the two genuine limits in the paper.
(4) The source-entropy-correlated recoverability ordering with empirical confirmation (Section 7): the inverse entropy–recovery relationship over 30 per-(axis, slot) cells ($r = -0.83$, Spearman $\rho = -0.91$) on a float-epsilon-corrected retrained checkpoint, with a companion decoder and an independent baseline landing at the same ceiling on the categorical command head (with a Miller–Madow bias note and reported per-cell counts).
(5) A resolved account of the numeric target-encoding artifact (Section 8): the encoding was corrected, the model retrained 5-fold, and the entropy/accuracy re-derived with a consistent encoding; the prior slot-5 three-regime structure is shown to have been an artifact now removed. A vocabulary-cardinality study (Section 9) and an audit-leverage corollary (Section 10) complete the contribution.

Scope statement.

Every quantitative result is conditional on a single corpus: CAM-authored toolpaths for a single desktop CNC platform (Bantam Tools desktop mill), a single workpiece-material class, nine machining operation classes, a closed V8 token vocabulary, and a 4 Hz envelope-extracted sensor stream. The corpus holds spindle speed constant (`S12000`, a 0-bit deterministic field) and contains no arc-interpolation (I/J) variation, using R-notation arcs only. Whether the three-regime decomposition is platform-specific or a general property of CAM-generated toolpaths is the principal open question (Section 11); a second corpus through a different CAM postprocessor or a second machine is the registered next step. Until then the ordering is scoped to this corpus, not claimed as general.

The remainder is organized as follows. Section 2 reviews the information-theory and CAM-generation literature; Section 3 describes the G-code source and its quantization spec; Section 4 computes decoder-free source entropy; Section 5 develops the three-regime decomposition; Section 6 owns the 4 Hz floor; Section 7 states and confirms the coupled ceiling; Section 8 walks back the encoding artifact; Section 9 reports the vocabulary-cardinality study; Section 10 draws the audit-leverage corollary; Section 11 states limitations; Section 12 concludes.

2. Related Work

We review three bodies of work: the information-theory canon that supplies the recoverability bound (Section 2.1); CAM toolpath generation, which supplies the mechanism behind the entropy structure (Section 2.2); and side-channel reconstruction of manufacturing instructions, the closest empirical analog (Section 2.3).

2.1. Information Theory of a Structured Source

Shannon's mathematical theory of communication [5] establishes entropy as the fundamental measure of source uncertainty, and his prediction-of-printed-English study [4] demonstrates the per-symbol entropy of a structured discrete source — the direct conceptual ancestor of the per-digit source-entropy measurement developed here. Fano's inequality [2] lower-bounds the error probability of estimating a symbol from an observation by its *conditional* entropy $H(X \mid \text{obs})$; we use it only to motivate the two endpoints and report the marginal $H(X)$ as an empirical ordering, not a Fano bound (Section 1, "Why this is an information-theoretic question"). Feder and Merhav [6] sharpen the entropy–error-probability relationship, and Cover and Thomas [3] give the modern treatment of entropy, the asymptotic equipartition property, and rate–distortion theory [7] that frames lossy coordinate recovery. Because we estimate entropies from finite per-cell symbol counts, the plug-in estimator carries a known downward bias of order $(K-1)/(2n \ln 2)$ for a K-symbol alphabet and n observations [3]; we quantify the Miller–Madow bias correction [8,9] — negligible (≤ 0.004 bits) on the load-bearing X/Y/Z slots and material only for the rarest small-n op-class cells — and report per-cell n

throughout, so the reported plug-in estimates carry their own bias bound (Section 4). The sub-Nyquist floor — one of the two genuine limits here — rests on the Nyquist–Shannon sampling theorem [10,11]: a transient shorter than one sample period cannot be resolved, which is precisely the situation for the majority of G-code blocks at 4 Hz (Section 6). Finally, the digit-slot structure of coordinates is a scalar-quantization phenomenon in the Lloyd–Max sense [12,13]: the CAM postprocessor's coordinate grid is the quantizer, and the entropy of each digit-slot is the entropy of the corresponding quantization residual. We borrow only standard results; the contribution is their application to the CNC toolpath source.

2.2. CAM Toolpath Generation and Numerical Emission

The entropy structure we measure is a fingerprint of how the toolpath was generated. Traditional facing and pocketing strategies emit coordinates on a geometrically constrained set (workpiece edges, fixed step-overs), whereas adaptive high-speed-machining (HSM) and trochoidal strategies compute continuously curving engagement-controlled paths whose coordinates are effectively continuous-valued at the postprocessor's emission precision; Otkur and Lazoglu [14] model the trochoidal toolpath and its cutting mechanics, the canonical instance of an optimizer emitting effectively-continuous fractional coordinates. Pocketing toolpaths generated from medial-axis / Voronoi decompositions, treated comprehensively by Held [15], fall between, mixing repeated wall-follow waypoints with optimizer-generated infill. The numerical emission itself — how many decimals a postprocessor writes, and whether trailing digits encode geometry or floating-point residue — is governed by postprocessor and CNC-system design [16], while feedrate and spline interpolation [17] set the curvature-dependent spacing of emitted points and hence the fine-digit statistics. The machine-tool positioning accuracy and repeatability that bound the meaningful coordinate precision are standardized by ISO 230-2 [18]; for a desktop platform this resolution is far coarser than the four decimals the postprocessor writes, so the finest emitted digits lie below the machine's executional resolution. Their near-uniform source distribution, however, arises on the CAM side — the engagement-controlled optimizer emits floating-point coordinate residuals — not from controller rounding, which would instead snap coordinates onto a coarse grid and *lower* the fine-digit entropy. The STEP-NC data model (ISO 14649 [19]) standardizes the upstream feature description, but the down-stream coordinate stream a controller executes is what a sensor observes, and its statistics are what we quantify. To our knowledge the source entropy of this stream, decomposed by generation strategy, has not previously been reported.

2.3. Side-Channel Reconstruction of Manufacturing Instructions

The closest empirical analog is side-channel reconstruction of additive-manufacturing toolpaths. Al Faruque *et al.* [20] recovered FDM print geometry from acoustic emission; Song *et al.* [21] from a smartphone microphone; Chhetri *et al.* [22] extended the idea to attack detection; later work uses optical [23] and acoustic-magnetic [24] channels. Most relevantly, Chhetri *et al.* [25] and Faezi *et al.* [26] compute *mutual information* between FDM G-code parameters and high-rate (kHz–MHz) emissions to characterize per-axis bits-of-leakage. Our framing differs along three axes. First, the host domain is subtractive CNC milling, a harder signal environment (simultaneous multi-axis motion, structural vibration, high-energy chip ejection). Second, the entropy object is not channel mutual information but the per-field *source* entropy of the CNC corpus itself; unlike [25,26] we do not estimate $H(X \mid \text{obs})$ or the channel mutual information, so we present the source entropy as a recoverability *ordering* confirmed empirically rather than a tight channel-agnostic bound. Third, we decompose the fine-digit ceiling by toolpath-generation strategy and derive a defense-in-depth audit corollary. The sensing regime is also categorically lower-rate: a 4 Hz multi-modal aggregate envelope versus raw kHz–MHz acoustics, which makes the sub-Nyquist floor a first-class, genuine limit rather than a footnote.

3. The Source: CNC G-Code as a Structured Symbol Stream

Corpus.

The source-entropy apparatus of Section 4 reads nine authored CAM G-code toolpath files spanning three families — facing (`face`, `face075`, `face150`), pocketing (`pocket`, `pocket075`, `pocket150`), and adaptive HSM (`adaptive`, `adaptive075`, `adaptive150`), the `075`/`150` suffixes denoting feed/speed variants — on a single desktop CNC platform. (The companion decoder [1] is trained on the windowed sensor corpus, whose nine operation-class labels include the damage and `150025` variants used in the per-(head, op-class) matrix of Section 7; the decoder-free entropy figures in this paper are computed on the raw source files named here.) The sensor-side provenance and the multi-modal denoising encoder are described in the companion contributions [1,27], which this paper takes as fixed. The toolpaths run in inch mode (1 in = 25.4 mm); arcs use R notation exclusively (no I/J), and spindle speed is held constant at a single commanded value (`S12000`) across all nine files, so its source entropy is identically 0 bits (a deterministic field, the cleanest endpoint of the framework).

Quantization spec — two distinct grids.

The analysis turns on a clean separation between the *source* and its *tokenized representation*, so we fix the spec carefully. The CAM postprocessor writes coordinate *fields* with **four** decimal places, and the corpus is genuinely four-decimal: of the X/Y/Z coordinate tokens that carry a fractional part, 11,173 are written to four decimals (*e.g.* `X3.3121`, `Y1.4848`) versus 1,456 to three (both counted directly on the nine `data_clean/*.gcode` toolpath files), and the ten-thousandths digit varies near-uniformly in the source (source entropy $\sim$3.31 bits on X/Y/Z; Section 4). The source's intrinsic precision floor therefore sits at the *hundredths–thousandths* (10^{-2}–10^{-3} inch, 254–25.4 µm), where the digit distribution reaches the uniform ceiling and is unrecoverable from the low-rate stream; the ten-thousandths is equally uniform in the source and carries no special status *at the source*.

Separately, the V8 grammar tokenizer that maps coordinates onto the discrete vocabulary (K = 2,418 numeric buckets, [28]) quantizes each axis to a per-axis precision grid: 0.001 for X/Y/Z, 0.0001 for R, and 1.0 for F (the spec is in `src/miracle/dataset/preprocessing_utils.py`, key `DEFAULT_PRECISION`). The 0.001 X/Y/Z grid *discards* the source ten-thousandths: a value `X3.3121` tokenizes to `X3.312`, so the 10^{-4} slot of the tokenized target is identically a constant (0 source bits) even though the source digit was near-uniform. R, tokenized on the 0.0001 grid, retains all four decimals. **The X/Y/Z 10^{-4} "structural zero" is thus a tokenizer representation choice, not a property of the source or the sensor.** We present both quantities side by side (Table 1); the only slot the tokenizer grid drives to a constant (0 bits) is the X/Y/Z ten-thousandths (the F tenths likewise collapse under the integer 1.0 grid, but F is omitted from Table 1 for thin support), and elsewhere source and tokenized agree to within $\leq$0.01 bits.

For entropy analysis we decompose every coordinate value v onto a fixed six-slot decimal grid $[10^1, 10^0, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}]$, indexed slot 0 (tens), slot 1 (units), slot 2 (tenths), slot 3 (hundredths, 10^{-2}), slot 4 (thousandths, 10^{-3}), and slot 5 (ten-thousandths, 10^{-4}). Slot s carries the digit at the corresponding decimal place of $|v|$; the sign is treated as a separate one-bit channel (Section 4). This is the Lloyd–Max view [12]: each slot is a uniform quantizer of one decade, and its entropy is the entropy of the digit it carries across the corpus. We compute slot entropy on both the raw four-decimal source ("source") and after applying the per-axis tokenizer grid ("tokenized"); the vocabulary-cardinality study (Section 9) varies the grammar's bucket count and is otherwise taken as given.

4. Per-Field Source Entropy, Decoder-Free

This section establishes the central object of the paper — the per-field source entropy of the toolpath coordinates — and does so with no model in the loop, so that the entropy structure is demonstrably a property of the source files.

Apparatus.

We parse the raw CAM G-code text of the nine authored toolpath files (6,051 code lines) with the regular expression `([XYZIJKRF])(-?\d+\.?\d*)`, strip controller comments, and extract 5,646 X, 5,521 Y, 1,707 Z, 431 R, and 16 F coordinate tokens. The synthetic vocabulary-coverage file `data/vocab_corpus.gcode` is deliberately excluded, as it enumerates bucket values for tokenizer coverage rather than representing an authored toolpath; including it would bias the source-entropy estimate toward uniform. Each value is decomposed onto the six-slot grid of Section 3, and we compute the Shannon entropy $H_s = -\sum_{d=0}^{9} p_s(d) \log_2 p_s(d)$ of the empirical digit distribution at each slot s, per axis and per operation class, plus a one-bit sign entropy per axis. The slot grid, regex, digit-encoding, and entropy routine are matched verbatim to the model-side analysis of the companion work so the two are directly comparable; the only intentional differences are that the input is raw source text (not tokenized windows), operation class is the source-file basename, and comments are stripped before extraction. Crucially, this apparatus never reads the NPZ windows, the tokenizer, or the decoder target encoding. The producer is released as `scripts/analysis/raw_gcode_source_entropy.py` and its output as `audit/raw_gcode_source_entropy.json`.

Entropy-estimator bias.

The plug-in entropy H_s underestimates the true entropy by approximately $(K_s - 1)/(2n_s \ln 2)$ bits for an alphabet of K_s observed symbols and n_s observations [3,8]. We apply the Miller–Madow correction $\hat{H}_s^{\mathrm{MM}} = H_s + (\hat{K}_s - 1)/(2n_s \ln 2)$, where $\hat{K}_s$ is the number of digits with non-zero empirical mass at slot s. On the load-bearing X/Y/Z digit slots ($K_s = 10$, n_s ranging from 1,707 on Z to 5,646 on X) the correction is ≤ 0.004 bits (≤ 0.0012 bits on X/Y, 0.004 bits on the smaller Z slot) and changes no reported value; it matters only for the small-n op-class cells, where we report its magnitude in-text and release the per-cell symbol counts n from which it is reproducible (`audit/per_head_per_opclass_entropy_5fold.json`, key `per_fold_observation_counts`); cells with $n < 30$ are flagged (Figure 1). The released JSON stores the uncorrected plug-in entropies and these counts, not the corrected values. All per-axis and per-slot entropies in this section are well-supported ($n > 700$); the only $n < 30$ cells are individual (head, op-class) pairs in the rarest operation classes, which we never treat as load-bearing.

The entropy structure is a property of the source.

Five structural claims reproduce decoder-free from the raw source:

(i) **High-order digits are near-deterministic.** Slot-0 (tens) entropy is 0.000 bits for X, Y, Z, and R — every coordinate satisfies $|v| < 10$ in this small-part corpus — identical between raw source and the model-side estimate.

(ii) **Fine decimals plateau near the uniform ceiling across the hundredths–thousandths.** For X/Y the 10^{-1} through 10^{-4} slots rise to 3.20–3.31 bits and *plateau* from the hundredths (10^{-2}, 3.31 bits) onward, empirically indistinguishable from uniform over $\{0, \ldots, 9\}$ ($H_{\max} = \log_2 10 \approx 3.32$ bits). The source's intrinsic precision floor is therefore the hundredths–thousandths band ($\sim$3.3 bits), where the source carries no concentrated structure for a sensor to recover.

(iii) **The X/Y/Z ten-thousandths is uniform in the source but discarded by the tokenizer.** In the four-decimal source the 10^{-4} slot is near-uniform on X/Y/Z (source entropy $\sim$3.31 bits), identical to the thousandths; its appearance as a structural zero is a tokenizer representation artifact (the 0.001 grid; canonical statement in Table 1 and below). This source-vs-tokenized split is the load-bearing correction to the prior draft, which conflated the tokenizer's representable floor with a property of the three-decimal-coordinate "corpus".

(iv) **Three operation-class regimes at the thousandths.** Per-op-family X-axis source entropy at the thousandths (10^{-3}, slot 4) tiers as 2.10 bits (facing), 3.32 bits (pocketing), and 3.31 bits (adaptive) — a clean low/(high≈high) split (Section 5). Facing's coordinate-repeat structure is real source structure and lives at the thousandths.

(v) **R is a low-entropy discrete vocabulary, retained at all four decimals.** Arc radii are positive (sign entropy 0.00 bits) with slot entropies in the 0.67–2.51-bit range, below the X/Y/Z ceiling, reflecting a small repeated CAM arc-radius set. Because R is tokenized on the finer 0.0001 grid, it retains its 10^{-4} digit (source = tokenized = 2.51 bits) and is recoverable across all slots — the one axis where the ten-thousandths is genuinely represented.

Source versus tokenized: the only collapsed slot is the X/Y/Z ten-thousandths.

Table 1 reports the full per-(axis, slot) entropy on both the four-decimal raw source and the per-axis tokenizer grid (`audit/source_vs_tokenized_entropy.json`). The two columns *agree to within* $\leq$ 0.01 *bits at every slot except the X/Y/Z ten-thousandths (and the F tenths, trivially discarded by the integer* 1.0 *grid; F omitted for thin support)*, where the 0.001 tokenizer grid collapses a near-uniform source digit (3.31 bits) to a constant (0.00 bits); the sub-0.01-bit shifts at the Z tenths/hundredths are grid-snap rounding, not structure. The hundredths and thousandths are at the uniform ceiling on both X and Y (3.31–3.32 bits, within 0.02 bits of $\log_2 10$); Z lifts off its tenths plinth (0.64 bits, cutting-depth steps) to the ceiling by the hundredths. R, tokenized on the finer 0.0001 grid, agrees source-to-tokenized at all six slots, including the ten-thousandths (2.51 bits). This table is the precise statement of the recoverability limit: the genuine, sensor-independent floor is the hundredths–thousandths plateau ($\sim$3.3 bits, source = tokenized), while the X/Y/Z 10^{-4} "structural zero" is a representation artifact of the 0.001 grid, recovered trivially as the constant the tokenizer makes it.

Table 1. Per-(axis, slot) Shannon entropy (bits) on the raw four-decimal *source* versus the V8-*tokenized* target (per-axis precision grid: X/Y/Z = 0.001, R = 0.0001, F = 1.0). Source from `audit/source_vs_tokenized_entropy.json`, key `per_axis_slot`. Slot indices: 0 = 10^1 tens, 1 = 10^0 units, 2 = 10^{-1} tenths, 3 = 10^{-2} hundredths, 4 = 10^{-3} thousandths, 5 = 10^{-4} ten-thousandths. **Source and tokenized agree to within** $\leq$ 0.01 **bits at every slot except the X/Y/Z ten-thousandths** (**bold**), the only slot the 0.001 grid drives to a constant ($\sim$ 3.31 $\to$ 0 bits); the residual < 0.01-bit Z tenths/hundredths differences are grid-snap rounding. R retains its ten-thousandths because it is tokenized on the finer 0.0001 grid.

Axis		10^1 (tens)	10^0 (units)	10^{-1} (tenths)	10^{-2} (hund.)	10^{-3} (thous.)	10^{-4} (ten-th.)
X (0.001)	source	0.00	1.72	3.21	3.31	3.30	3.31
	tokenized	0.00	1.72	3.21	3.31	3.30	**0.00**
Y (0.001)	source	0.00	1.68	3.20	3.31	3.32	3.31
	tokenized	0.00	1.68	3.20	3.31	3.32	**0.00**
Z (0.001)	source	0.00	0.00	0.64	3.23	3.31	3.31
	tokenized	0.00	0.00	0.65	3.24	3.31	**0.00**
R (0.0001)	source	0.00	0.09	0.67	2.38	2.47	2.51
	tokenized	0.00	0.09	0.67	2.38	2.47	2.51

Raw-source versus model-side agreement.

Table 2 compares the per-axis hundredths/thousandths source entropy against the corrected, encoding-consistent model-side estimate, confirming that the central object reproduces decoder-free. The load-bearing X/Y/Z axes sit at 3.24–3.32 bits at the thousandths (source 3.30–3.32, model-side 3.24–3.31), within $\sim$ 0.07 bits of the uniform ceiling; the model-side estimate (computed on the patched-encoder targets, hence on the 0.001-tokenized values) agrees at the hundredths/thousandths and reports the X/Y/Z ten-thousandths as the tokenizer's structural zero. This is the salami-defense pivot: the paper's central quantity — a fine-digit ceiling at the hundredths–thousandths — is a property of the source, and the model-side estimate agrees with it once the encoding is consistent (Section 8).

Table 2. Per-axis entropy at the two finest slots, source (four-decimal raw, `audit/source_vs_tokenized_entropy.json`) versus the corrected model-side estimate (`audit/patched_numeric_analysis.json`, key `corrected_per_axis_slot_entropy`; computed on 0.001-tokenized X/Y/Z targets). The thousandths (10^{-3}, slot 4) is the uniform-noise ceiling on X/Y/Z (3.24–3.32 bits, source = model-side). The ten-thousandths (10^{-4}, slot 5) is near-uniform in the four-decimal source on X/Y/Z but a structural zero in the 0.001-tokenized model target; R retains it on both ($\approx$ 2.0–2.5 bits) because it is tokenized at 0.0001.

	10^{-3} thousandths (slot 4)		10^{-4} ten-thousandths (slot 5)	
Axis	source	model-side (tok.)	source	model-side (tok.)
X	3.30	3.236	3.31	0.00
Y	3.32	3.295	3.31	0.00
Z	3.31	3.310	3.31	0.00
R	2.47	2.327	2.51	1.989
F	0.00	0.00	0.00	0.00

Per-axis sign entropy.

The motion-sign of each axis is a one-bit source channel, and its entropy is sharply axis-asymmetric (Table 3, raw source). Y is prior-deterministic ($H_Y^{\mathrm{sign}} = 0.11$ bits, 99% positive) and X near-prior (0.23 bits); only Z carries a genuine binary load (1.10 bits, 49% positive / 49% negative), splitting between rapids above the stock (positive) and cuts into it (negative). R and F are uniformly positive (0.00 bits). The model-side estimate over the 5-fold test split gives the same ordering ($H_X^{\mathrm{sign}} = 0.38$, $H_Y^{\mathrm{sign}} = 0.08$, $H_Z^{\mathrm{sign}} = 1.04$ bits); the two populations differ in absolute value on X/Y because the windowed split re-weights the rare negative moves, but the qualitative claim — only Z's sign carries recoverable information — is invariant. This explains the per-axis sign-recovery asymmetry observed by the companion decoder [1] without reference to the decoder: a head cannot recover information a deterministic source does not contain.

Table 3. Per-axis motion-sign source entropy. Raw-source columns are decoder-free (`audit/raw_gcode_source_entropy.json`, key `per_axis_sign`); the model-side column is the 5-fold test-split estimate (`audit/digit_entropy.json`, key `per_axis.{X,Y,Z}.sign`) and is the 3-way (positive/negative/zero) sign entropy. Only Z's sign is a non-trivial binary classification; X and Y are prior-dominated and R/F are uniformly positive. The two populations re-weight the rare negative X/Y moves differently but agree on the ordering.

Axis	Raw H^{sign} (bits)	Positive frac. (raw)	Model-side H^{sign} (bits)	Interpretation
X	0.23	0.964	0.38	near-prior
Y	0.11	0.987	0.08	prior-deterministic
Z	1.10	0.494	1.04	genuine binary load
R	0.00	1.000	0.00	uniformly positive
F	0.00	1.000	0.00	uniformly positive

Per-head and per-position entropy across all five folds.

Beyond the per-(axis, slot) decomposition, we compute the per-(head, op-class), per-head global, and per-position source entropy aggregated across all five training folds and report each as a 5-fold mean $\pm$ s.d. A dedicated recompute confirmed that the 5-fold aggregate equals the fold-1 estimate to within reporting precision: the per-head op-class-conditional mean entropy (the average of $H(\text{head} \mid \text{op-class})$ over the nine classes) shifts by at most $+0.030$ bits between fold-1 and the 5-fold mean (command: $1.468 \rightarrow 1.498 \pm 0.032$ bits; all other heads ≤ 0.015 bits; digit unchanged to four decimals), and the largest single (head, op-class) cell shift is $+0.12$ bits in a low-count class (command $\times$ `adaptive150025`, $n \approx$ 36–45 per fold; the genuinely under-supported cells, e.g.

command $\times$ `damageadaptive` at $n = 9$, are hatched in Figure 1). The fold-1-versus-5-fold provenance question raised in prior review is therefore resolved as cosmetic. These op-class-conditional means are distinct from—and necessarily no larger than—the *marginal* per-head source entropies reported in Table 4 (`audit/per_head_source_entropy_5fold.json`): for the command head the marginal entropy is 1.67 bits against the 1.50-bit op-class-conditional mean quoted here, the difference being exactly the between-op-class component that conditioning removes. Figure 1 maps the per-(head, op-class) ceiling (cells with $n < 30$ are hatched), Figure 2 the per-position swing, and Figure 3 the per-toolpath decomposition. As Section 7 shows, aggregating to five folds leaves the entropy–accuracy ordering intact.

Table 4. Per-head source entropy versus measured recovery accuracy (patched encoder). Empirical entropy H_{emp} is the 5-fold mean $\pm$ sample s.d. of the per-fold training-target distribution, recomputed for every head from the five fold NPZs under one decomposition (`audit/per_head_source_entropy_5fold.json`); the per-head 5-fold mean equals the fold-1 estimate to within $\leq$ 0.015 bits on every head (`h_emp_delta_vs_fold1` in the JSON), so the earlier fold-provenance inconsistency is resolved. $H_{\max} = \log_2 |\mathcal{V}_{\mathrm{head}}|$ is the uniform-distribution ceiling over the canonical head vocabulary. Accuracies are 5-fold test means on the float-epsilon-corrected retrain. The categorical heads (type, command, parameter-type, sign) are per-position classifiers on the encoder memory and are teacher-forced/autoregressive *invariant* (AR = TF, marked *); only the token and digit heads consume the autoregressive stream and collapse under AR. Recovery tracks source entropy across heads: low-entropy heads (param-type, sign) recover near-perfectly, while the pooled digit head (2.64 bits, finest decimals at the $\log_2 10$ ceiling) drops to 0.570 TF / 0.10 AR. The TF$\to$AR collapse on token and digit is the exposure-bias signature characterised in Section 7, not a source-entropy effect.

Head	Vocab $\lvert\mathcal{V}\rvert$	$H_{\max}$ (bits)	H_{emp} (bits)	TF acc.	AR acc.	Acc. gap (1–TF)
Token	2,418	11.24	6.97 ± 0.00	0.774	0.215	0.226
Type	4	2.00	1.24 ± 0.01	0.983	0.983*	0.017
Command	6	2.58	1.67 ± 0.01	0.891	0.891*	0.109
Param-type	6	2.58	1.74 ± 0.00	0.993	0.993*	0.007
Sign (pooled)	2	1.00	0.33 ± 0.00	0.989	0.989*	0.011
Digit (pooled)	10	3.32	2.64 ± 0.00	0.570	0.104	0.430

**Categorical heads are per-position classifiers on the encoder memory and do not consume the autoregressive token stream, so their AR accuracy equals their TF accuracy. The accuracy gap is $1.0 -$ TF accuracy (distinct from the $\log_2 \lvert\mathcal{V}\rvert$ entropy ceiling). The command TF accuracy reported here (0.891) is the patched-retrain value; the released-checkpoint value (0.888, used in the seq2seq comparison of Table 5) differs by <1.5 pp, consistent with the categorical heads being TF/AR- and checkpoint-invariant under the patch. All H_{emp} are 5-fold mean $\pm$ s.d. from `audit/per_head_source_entropy_5fold.json`; the sign-pooled value is the n-weighted mean across sign-X/Y/Z/F (sign-S has $n \approx 0$); this is the 3-way (positive/negative/zero) entropy and equals 0.33 bits as printed (the renormalised positive-vs-negative entropy is lower). The pooled digit entropy (2.64 bits) is lower than the per-slot fine-decimal ceiling ($\sim$3.3 bits, Table 1) because it pools the near-deterministic integer slots with the near-uniform fractional slots.*

5. Three Operation-Class Regimes

The fine-digit entropy ceiling is not uniform across the corpus; at the thousandths (10^{-3}, slot 4) it stratifies into three regimes whose boundaries are dictated by CAM toolpath-generation strategy, not by the recoverer. We locate the stratification at the thousandths — the deepest slot the source genuinely varies that is also retained by the tokenizer — and report the SOURCE entropy, which is the property of interest here. The prior draft placed the regimes at slot-5 (10^{-4}); the float-epsilon correction (Section 8) shows that placement was an encoding artifact. The genuinely tokenizer-discarded ten-thousandths is recovered near-perfectly for every op-class (0.97–1.00; Figure 4) precisely because the 0.001 grid makes it a constant, not because it is fine-digit signal. Figure 5 and Figure 6 give the full slot$\times$op-class source surfaces.

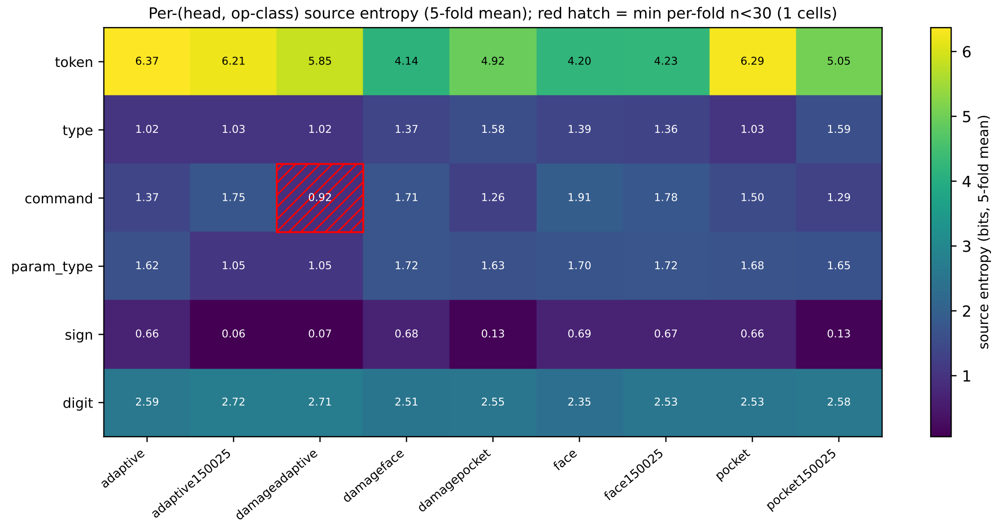

Figure 1. Per-head per-operation-class empirical source entropy (bits) on the training corpus, aggregated over five folds. The same spatial pattern is mirrored by measured recovery (Figure 9): low-entropy cells (facing-class commands, small-vocabulary parameter types) admit high recovery; high-entropy cells (adaptive-class numeric digits) saturate at the information-theoretic ceiling.

(i) Adaptive HSM — the uniform-noise ceiling.

Adaptive HSM toolpaths (the raw apparatus files `adaptive`, `adaptive075`, `adaptive150`) produce the canonical CAM-interpolation signature: per-axis thousandths *source* entropy $\approx$ 3.31 bits (X) with low mass on any single digit, the fingerprint of an engagement-controlled optimizer emitting floating-point residuals rather than round-ending coordinates. Adaptive X/Y decimals (tenths through thousandths) all sit at 3.1–3.3 bits, and Z lifts off a tenths plinth (0.61 bits for the adaptive family, cutting-depth steps; 0.64 bits pooled across families) to $\approx$ 3.26–3.31 bits at the hundredths and thousandths. These digits are at the uniform ceiling: their source carries no concentrated structure, so a sensor channel that does not directly observe the coordinate leaves a recoverer at the prior.

(ii) Facing — geometric coordinate repetition.

Facing operations (the raw apparatus files `face`, `face075`, `face150`) have *thousandths* source entropy $\approx$ 2.10 bits (X, $n = 324$) with substantial mass on a few digits — the mechanism is geometric *coordinate repetition*, not a precision grid. Specific X values dominate the `face` class: the facing-pass entry/exit coordinates at the workpiece left and right edges (*e.g.* $X = -0.1254$ at 27%, $X = 3.5354$ at 21%, with $X = 3.5725$ and $X = -0.1625$ at $\sim$7% each — together $\sim$62% of facing X-coordinates) recur and concentrate the thousandths digit. Low entropy here is genuine source structure, but it is workpiece-geometry-specific and would not transfer to a different part. (The prior draft's "96.5% of face X from four overruns at slot-5" figure was an artifact of the float-epsilon encoding, which folded terminal zeros into a spurious digit-9 mass at the ten-thousandths; the coordinate-repeat structure survives the correction but is correctly located at the thousandths.)

(iii) Pocketing — not an intermediate, but at the ceiling.

Pocket-style operations (`pocket`, `pocket075`, `pocket150`) sit at the ceiling alongside adaptive: X thousandths source entropy 3.32 bits, essentially tied with adaptive (3.31 bits) and well above facing (2.10 bits). The earlier "intermediate" framing was a consequence of the mislocated slot; under the corrected analysis the clean split is facing (low) versus pocketing$\approx$adaptive (at the ceiling). Pocket walls do contribute repeated coordinates (*e.g.* `Y-0.071` recurring), but on the X thousandths the optimizer-generated infill dominates.

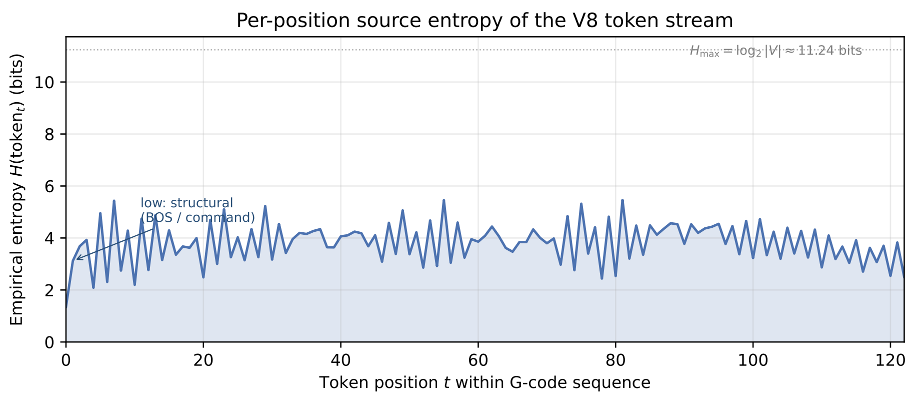

Figure 2. Per-position empirical source entropy of the V8 token stream. Structural positions (BOS, command-letter, axis-letter) sit near 1–2 bits; numeric-digit positions rise to $\sim$ 5–5.5 bits, well below the vocabulary ceiling $H_{\text{max}} = \log_2 |V| \approx 11.24$ bits. The $\sim 3\times$ within-line swing is the within-line analogue of the per-(head, op-class) ceiling. Regenerated from `audit/per_position_token_entropy_5fold.json`.

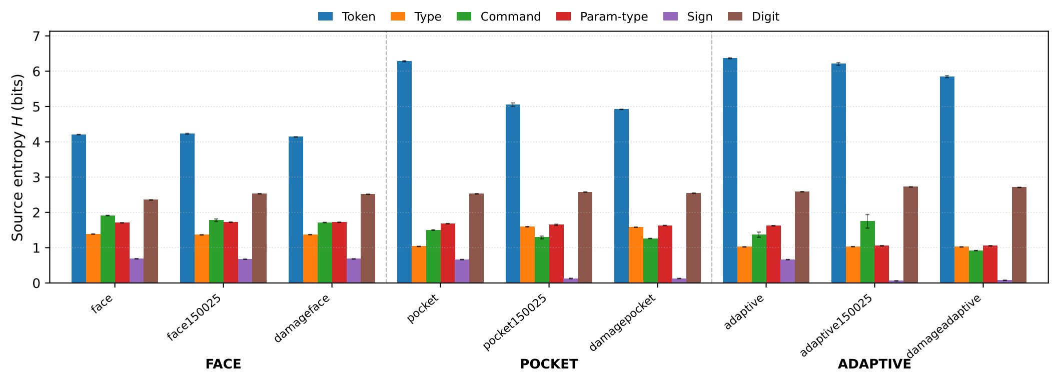

Figure 3. Per-toolpath empirical source entropy decomposed by prediction head across the nine operation classes, grouped by toolpath family (face / pocket / adaptive). The token head carries the largest absolute entropy; the digit head shows the cleanest facing-vs-adaptive separation: facing operations carry $\sim$2.4 bits of digit-head entropy (overruns repeat workpiece-edge coordinates) while adaptive operations carry $\sim$2.6 bits (near-uniform fractional digits). Sign and parameter-type heads are nearly toolpath-invariant. Regenerated from `audit/per_head_per_opclass_entropy_5fold.json`.

Per-axis specificity.

The regime ordering also carries a per-axis structure (Figure 6). At the thousandths, facing X drops to ≈ 2.1 bits (the workpiece-edge repeats), while Y stays near the ceiling (facing passes traverse Y on every cut). Z is the per-axis exception in the other direction: its sign and tenths carry concentrated cutting-depth structure (0.64 bits at the tenths, recovered through the hundredths at 0.91), and only its hundredths-and-finer approaches the ceiling. Recoverable source structure is therefore both toolpath- and axis-specific: X carries facing-class signal, Z carries cutting-depth signal.

6. The Physical Floor: 4 Hz Sub-Nyquist Aliasing

The source-entropy ceiling of Sections 4–5 bounds recoverability from above given a perfectly resolving sensor. A second, independent bound comes from the sensing channel itself: at a low sample rate, most G-code blocks execute faster than one sample period and cannot be resolved as distinct transients, so even a digit with low source entropy is unrecoverable if the block that carries it is temporally aliased.

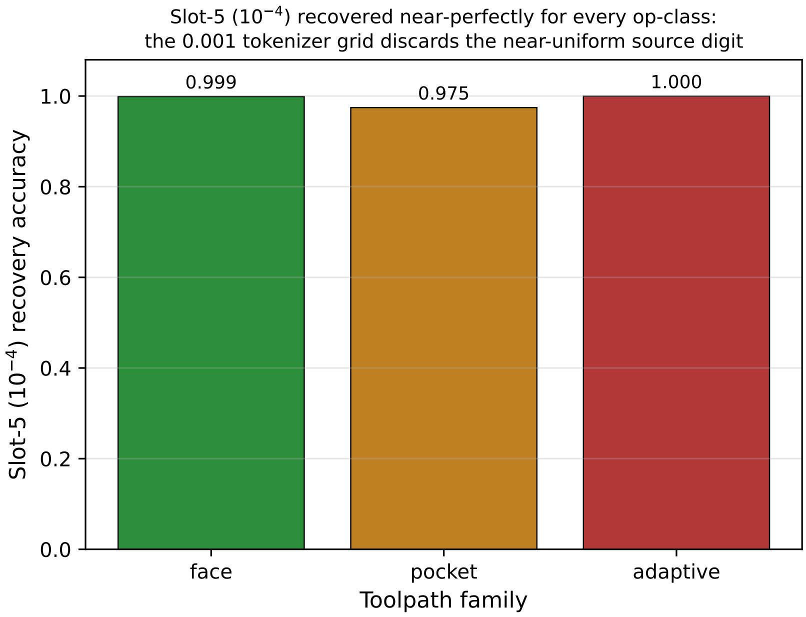

Figure 4. Slot-5 (10^{-4}-inch) recovery accuracy by toolpath family, patched encoder. The X/Y/Z ten-thousandths is discarded by the 0.001 V8 tokenizer grid (Table 1), so its tokenized target is a constant and all three families recover it near-perfectly (0.97–1.00) — a representation artifact, not fine-digit recovery. The op-class *source-entropy* stratification (facing 2.10 bits low / pocketing 3.32 ≈ adaptive 3.31 bits at the ceiling) lives at the thousandths, not here — see Figure 5.

Measured execution durations.

We measure the actual per-row execution durations across the 120-file aligned corpus (`scripts/analysis/gcode_row_durations.py`; `audit/gcode_row_durations.json`; 21,541 row-execution instances) against the released 4 Hz (0.25 s) sample period. The *median* row executes in a single sample (median 1.0 sample, 0.25 s), and 77.7% **of row-execution instances occupy** ≤ 1 **sample period** (≤ 0.25 s), with 83.1% ≤ 0.5 s and 86.3% ≤ 1 s. The duration distribution is heavy-tailed (mean 2.90 samples, p90 7 samples, max 181 samples), but the mass is concentrated at or below one sample.

The Nyquist consequence.

By the sampling theorem [10,11], a transient shorter than one sample period aliases: the 4 Hz stack observes the lower-frequency cutting-regime envelope, not the individual block. This is why categorical / regime-level structure (which operation family is cutting) is recoverable while per-instruction numeric precision is not, and the mechanism is physical, not merely an artifact of any recoverer's training dynamics. The sub-Nyquist floor and the source-entropy ceiling are therefore *coupled but distinct*: a block can be unrecoverable because its source digit is uniform (entropy ceiling), because the block is sub-sample (Nyquist floor), or both. Acoustic-emission and vibration tool-condition monitoring conventionally samples at ≥ 10 kHz [29]; the present stack sits deliberately at the low-rate, low-cost end, and the recoverability characterization must be read in that light. A higher-rate front end (a ≥ 44.1 kHz contact microphone, ≥ 1 kHz axis encoders) would lift the Nyquist floor but cannot lift the entropy ceiling on a uniform source digit.

7. The Recoverability Ceiling and Empirical Confirmation

We now confirm the recoverability ordering empirically on the float-epsilon-corrected, 5-fold-retrained checkpoint. **Claim (empirical ordering, not a tight bound):** per-field recovery accuracy of the CNC toolpath stream falls monotonically as the per-field marginal source entropy rises,

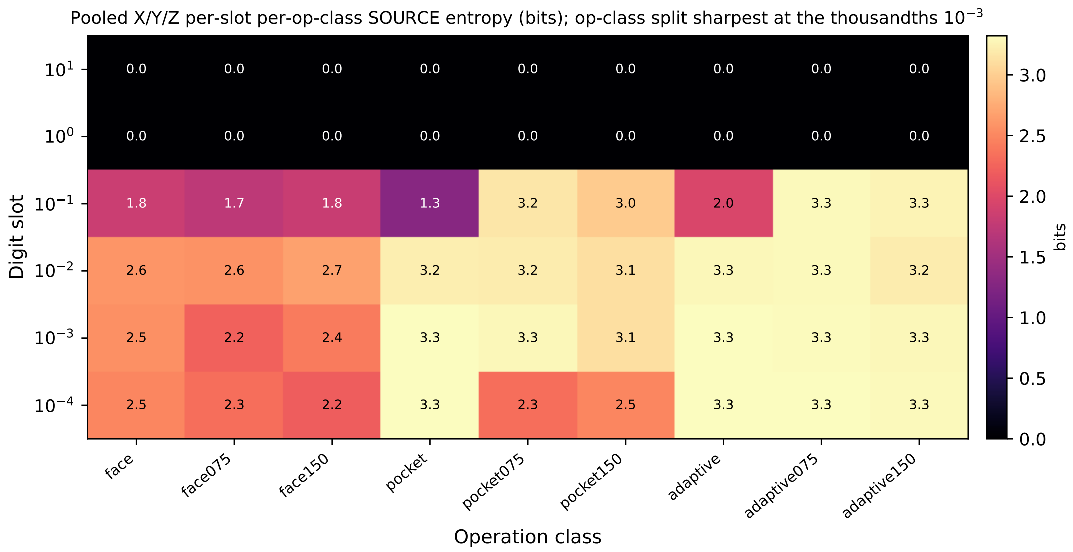

Figure 5. Pooled X/Y/Z per-digit-slot per-operation-class *source* entropy (bits). Rows: the six decimal slots (tens to ten-thousandths). Columns: nine operation classes. The tens/units rows are near-zero ($|v| < 10$ everywhere); the tenths through ten-thousandths plateau near the uniform ceiling ($\log_2 10 \approx 3.32$ bits) on the four-decimal source. The op-class stratification (facing below pocketing≈adaptive) is sharpest at the thousandths (10^{-3}). This is SOURCE entropy; the near-uniform X/Y/Z ten-thousandths shown here is discarded by the 0.001 tokenizer grid (Table 1). Regenerated from `audit/raw_gcode_source_entropy.json` (`figures/regenerate_source_apparatus_figures.py`).

and a learned recoverer lands at the ceiling on the heads that have one. This is an ordering keyed to $H(X)$, not a Fano bound (Section 2.1); the two genuine limits are the 0-bit slots and the 4 Hz floor (Section 6). The confirmation has three legs: the entropy–accuracy relationship at the cell level, the relationship at the head$\times$op-class level, and the convergence of two independent recoverers — which we restrict to the categorical command head.

Cell-level entropy–accuracy relationship (lead result, encoding-consistent).

The load-bearing statistic is the per-(axis, slot) inverse relationship over 30 per-(axis, slot) cells (five fields $\times$ six decimal slots; F enters at thin support, Section 11), re-derived on the patched encoder. Both axes of this relationship are the *tokenizer-representable* object: the entropy is the model-side tokenized per-(axis, slot) entropy of the patched targets (`audit/patched_numeric_analysis.json`, key `corrected_per_axis_slot_entropy`; it matches the raw-source "tokenized" rows of Table 1 to within population sampling on X/Y/Z, with R differing by $\sim$0.5 bits owing to its small windowed support) and the accuracy is the patched recovery accuracy, so the two are paired over exactly the digits the model can represent (the X/Y/Z 10^{-4} cell enters as the tokenizer's 0-bit constant, recovered near-perfectly). Under this consistent encoding: Pearson $r = -0.83$, Spearman $\rho = -0.91$ ($n = 30$; Fisher-z 95% CI $[-0.92, -0.67]$; `audit/patched_numeric_analysis.json`). The ordering is not an artifact of the deterministic endpoints: excluding the ten near-deterministic cells that recover trivially (accuracy $\geq$0.99 — the integer slots and the tokenizer-zeroed X/Y/Z ten-thousandths) it is $r = -0.77$ ($\rho = -0.87$, $n = 20$); excluding F's thin-support cells it is $r = -0.83$ ($n = 24$); and it survives controlling for per-cell support (partial $r = -0.80$ given log token count). It is *stronger* than the pre-patch $r = -0.68$ (which carried a known encoding inconsistency), although part of that gain is the X/Y/Z slot-5 ten-thousandths — a tokenizer-constant cell — recovering better under the patch rather than a change in the fine-digit relationship: restricted to the slots the source actually varies it attenuates to the $n = 20$ value above but holds. This is the honest empirical core of the paper. Wherever the tokenized digit distribution is highly non-uniform the recoverer recovers it accurately;

wherever it approaches the uniform-noise ceiling it cannot. Figure 8b shows the cell-level scatter; Figure 8a overlays the pooled per-slot accuracy (the bathtub) on the per-slot source entropy.

Pooled six-slot relationship (illustrative only).

At the pooled six-slot level the same anticorrelation holds and is visually stark, but we report it as *illustrative* rather than load-bearing: the pooled Pearson $r = -0.89$ and Spearman $\rho = -0.94$ are computed over only $n = 6$ points, and the Pearson coefficient assumes bivariate normality that cannot be checked at $n = 6$. The defensible inference rests on the 30-cell relationship above, not on the six pooled points.

The ordering is cross-head, not within-head.

Decomposing by head clarifies what drives the negative relationship. The categorical command head sits near its ceiling and is individually flat across operation classes (it has little entropy spread to track); the numeric/digit head carries the entropy spread and the accuracy spread. The overall negative ordering is therefore *cross-head* — it is the contrast between a near-ceiling categorical head and a high-entropy numeric head that produces the slope — not a within-head bound that holds head-by-head. Within the digit head alone the relationship is strong ($r = -0.90$, $n = 9$); the sign head alone is essentially flat ($r = -0.19$) and the command head is slightly *positive* ($r = +0.38$), so pooling sign+digit ($r = -0.87$, $n = 18$) recovers a between-head contrast rather than a within-head bound. The single genuinely within-head negative slope is the digit head. We state this explicitly so the ordering is not mistaken for an architecture-invariant per-head accuracy bound. (Methodological note: an earlier head$\times$op-class scatter silently fell back to a few hardcoded headline constants broadcast as flat bands; the relationship reported here uses live per-cell accuracies after that alignment bug was repaired.)

Two recoverers converge at the ceiling — on the categorical command head only.

The companion decoder [1] (frozen multi-modal encoder + grammar-constrained six-head decoder) and an independent from-scratch Transformer seq2seq baseline (no frozen encoder, no grammar mask, no structured heads) reach the same teacher-forced *command* accuracy (0.877 baseline vs. 0.888 decoder — released checkpoint; 0.891 on the patched retrain, the categorical heads being TF/AR-invariant and drifting ≤ 1.5 pp under the patch — well within fold variance; Table 5). On the categorical command head the convergence is genuine: the categorical signal in the sensor stream is recoverable by either architecture, so the ceiling there is sensor-side, not architectural. We explicitly do *not* extend this convergence claim to the numeric head: under teacher forcing the from-scratch baseline reports a numeric (digit) accuracy 18.4 pp *higher* than the decoder (0.754 vs. 0.570; Table 5), a consequence of the decoder's bucketized numeric vocabulary trading TF digit precision for grammar-mask tractability. The two architectures therefore do not converge on the numeric head, and we make no architecture-invariance claim there. Closing the residual numeric gap requires raising the sensor bandwidth (Section 6), not changing the recoverer; closing the small categorical gaps requires neither.

The bathtub as the inverse of entropy.

The pooled per-digit-slot recovery accuracy is bathtub-shaped (Table 6, Figure 7): slot 0 (tens) at 1.000; slot 1 (units) at 0.922 (an axis mix — Z near 1.0, X/Y near 0.90); the tenths–thousandths descending monotonically to the *thousandths* trough at 0.417 (the true precision floor); and slot 5 (ten-thousandths) recovering to 0.993. The per-axis decomposition (Figure 7b) shows the pooled bathtub averages structurally different per-axis profiles: X and Y descend monotonically to a thousandths trough (0.43/0.38); Z reaches 0.98 at the tenths (a shallow-depth, rapid-vs-cut bimodality — the tenths digit is 0 for rapids/retracts and 1 for the shallow $|Z| < 0.2$ in cuts, 0.64 bits source) and recovers through the hundredths (0.91, where the few repeated per-pass cut depths concentrate mass), troughing only at the thousandths (0.27); R is uniformly ≥ 0.87 across all slots (small arc-radius

Table 5. Head-to-head comparison of the from-scratch seq2seq baseline (8.64 M params, vanilla Transformer encoder–decoder, end-to-end trained, no frozen encoder, no grammar mask, no structured heads) against the main-paper headline configuration (frozen MM-DTAE encoder + grammar-masked decoder + 6-head structured output, ≈ 30 M decoder params). Mean $\pm$ s.d. across 5 file-level CV folds. Δ = Main $-$ Baseline, in accuracy points. Bold entries mark heads where one configuration wins by more than 5 pp. The teacher-forced (sweep-time) command accuracy is architecture-invariant (0.877 vs. 0.888, well within fold variance); the architectural lift concentrates on the autoregressive (deployment-true) regime, where the grammar mask, frozen pretrained encoder, and structured-head curriculum jointly recover +77 pp on command accuracy and +78 pp on parameter-type accuracy. The TF-numeric column carries a surprise in the opposite direction: the from-scratch baseline reports a +18.4 pp higher TF digit accuracy (0.754 vs. the patched decoder's 0.570), consistent with the main paper's discretization design (bucketization for downstream grammar-mask compatibility) sacrificing some TF-time digit precision. This numeric-head divergence is why the architecture-invariance claim is restricted to the categorical command head. The Main column mixes checkpoints by design: the categorical entries are the released checkpoint (TF/AR-invariant, ≤ 1.5 pp drift under the patch) while the numeric entry is the patched float-epsilon-corrected retrain (Section 8), the only tier the patch affects. The companion paper [1] reports this numeric gap as +16.9 pp against its released-checkpoint digit accuracy (0.585); the +18.4 pp stated here is against the patched decoder (0.570), the basis used throughout this paper.

Metric	Baseline TF	Main TF	Δ TF	Baseline AR	Main AR	Δ AR
Command	0.877 ± 0.031	0.888 ± 0.056	+0.011	0.122 ± 0.051	0.888 ± 0.056	+**0.766**
Token	0.865 ± 0.027	0.781 ± 0.022	−**0.084**	0.154 ± 0.071	0.215 ± 0.069	+**0.061**
Numeric	0.754 ± 0.053	0.570 ± 0.033	−**0.184**	0.100 ± 0.063	0.104 ± 0.032	+0.004
Param-type	0.979 ± 0.012	0.993 ± 0.004	+0.014	0.214 ± 0.082	0.993 ± 0.004	+**0.779**
Sequence	0.175 ± 0.026	–[†]	–[†]	0.029 ± 0.040	0.026 ± 0.019	−0.003

The from-scratch baseline is the most direct seq2seq null model: it shares the full-window target formulation, vocabulary, optimizer family, and 5-fold split with the main system but removes (a) the frozen MM-DTAE encoder, (b) the grammar mask, and (c) the auxiliary structured heads. The TF-command read demonstrates that an 8.64 M vanilla Transformer can already match the categorical-head ceiling under teacher forcing — the categorical signal in the sensor stream is recoverable by either architecture. The AR-command read reflects where *command identity is harvested, not which decoder is stronger: the baseline reads it from the collapsed autoregressive token stream* (12%, *the most-frequent-class prior), while the main system reads it from a non-autoregressive classifier on the encoder memory* (89%, *TF/AR-invariant). The lift is the value of bypassing autoregression for the categorical fields, not an autoregressive-decoder advantage. The TF-numeric reversal is consistent with the main paper's vocabulary design: bucketization of numeric ranges to a fixed precision is what makes the grammar mask tractable in the legacy-token head (Section 3), but it forfeits the few percentage points of TF digit accuracy that the baseline retains by predicting raw digit tokens without bucketization constraints. The AR-numeric column collapses both architectures to within a fold of each other, indicating that numeric AR is mode-collapse-limited (Section 7) regardless of the architectural envelope.*
[†] *Sequence TF is reported for the baseline only; the main paper treats TF sequence exact-match as ill-defined because under full teacher forcing the metric reduces to per-token TF accuracy collapsed to all-or-nothing (Section 7), so the Main-TF and Δ-TF Sequence cells are intentionally blank rather than missing data.*

vocabulary). The bathtub is the inverse image of the tokenized per-slot entropy profile of Section 4 — recovery is easy exactly where the tokenized target is concentrated. The slot-5 *upturn* is now correctly understood: the X/Y/Z ten-thousandths is recovered at 0.99 as the tokenizer constant it is made into (a representation artifact; Table 1), not an unrecoverable fine digit and not (as the prior draft had it) a coordinate-repetition-plus-encoding artifact. The deepest part of the bathtub is the thousandths, the finest digit the tokenizer retains on X/Y/Z and the genuine source precision floor.

Per-axis bit budget and what is reliably recoverable.

The negative ordering is half the story; the per-axis source-entropy map (Figure 11a, Figure 11b) also identifies the cells where recovery *is* structurally available, summarized as a per-axis bit budget in Figure 10. Coarse-resolution recovery is reliable across all primary axes: the tens digit of X/Y/Z is uniformly 0 and recovered at 100%; the units digit reaches $\geq 90\%$ on X/Y (mass on $\{0, 1\}$) and is identically 0 on Z; Z's tenths digit recovers at high accuracy from its shallow-depth rapid-vs-cut bimodality ($H = 0.64$ bits source, recovered 0.98: the tenths digit is 0 for rapids/retracts and 1 for

Table 6. Per-digit-position accuracy and source entropy on V8 full_window (5-fold mean ± std), **patched encoder**. All accuracies and entropies are computed on the float-epsilon-corrected, 5-fold-retrained checkpoint with a consistent round-encoding (Section 8; `audit/patched_numeric_analysis.json`). Slot 0 is the most-significant magnitude digit (tens place); slot 5 is the least-significant decimal (ten-thousandths). Entropy is Shannon entropy in bits of the empirical digit-value distribution across all supervised positions, pooled across axes (matching the accuracy denominator). The bathtub-shaped accuracy curve tracks per-slot source entropy: the trough is at the *thousandths* (10^{-3}, 0.42), the finest digit the corpus varies and the true precision floor; the ten-thousandths (10^{-4}) recovers to 0.99 because the 0.001 V8 tokenizer grid discards the near-uniform ($\sim$3.31-bit) source digit, making the tokenized target a structural zero, not because the fine digit is recovered. See Figure 8a for the overlay and Figure 11a for the per-axis decomposition.

Slot	Place	Accuracy (5-fold)	Entropy (bits)	n
0	10^1 (tens)	1.0000 ± 0.0001	0.007	79,345
1	10^0 (units)	0.9223 ± 0.0180	1.690	79,345
2	10^{-1} (tenths)	0.7478 ± 0.0253	3.079	79,345
3	10^{-2} (hundredths)	0.5465 ± 0.0269	3.301	79,345
4	10^{-3} (thousandths)	0.4165 ± 0.0239	3.287	79,345
5	10^{-4} (ten-thousandths)	0.9931 ± 0.0067	0.311	79,345

The reported $n = 79{,}345$ is the pooled across-axis numeric-position count (identical per slot because every axis contributes a digit at every slot); the per-axis composition differs by slot — in particular at the ten-thousandths, where X/Y/Z are tokenizer-constant and R is not. Per-axis decomposition is in `audit/patched_numeric_analysis.json`. The thousandths is the true information-theoretic precision floor (high-entropy quasi-uniform); the X/Y/Z ten-thousandths is a tokenizer-induced structural zero (the 0.001 grid discards the near-uniform source digit), while R on its 0.0001 grid retains it. The pooled slot-5 entropy (0.311 bits) is therefore nonzero only because the across-axis pool mixes the X/Y/Z tokenizer-zeroed ten-thousandths (0.00 bits) with R's retained 0.0001 digit ($\sim$2.0 bits tokenized); on X/Y/Z alone slot-5 is 0.00 bits (Table 1).

the shallow $|Z| < 0.2$ in cuts) and remains recoverable through the hundredths (0.91, where the few repeated per-pass cut depths concentrate); the per-pass depth *increments* themselves fall in the hundredths–thousandths near the uniform ceiling and are not recovered. R is recoverable across all six slots ($H = 0.09$–2.51 bits source) because the corpus uses a small discrete arc-radius set. This is consistent with the patched per-axis value MAE (in inches): Z 0.009 and R 0.021 are near-exact, while X 0.248 and Y 0.205 are bounded by the uniform hundredths–thousandths digits. The per-axis bit totals are marginal sums $\sum_s H(X_s)$, an *upper bound* on the joint per-axis coordinate entropy $H(X)$ (subadditivity; the gap is inter-slot redundancy, largest where coordinates repeat — e.g. R, whose joint coordinate-value entropy $\approx$3.2 bits sits well below its $\approx$8.1-bit marginal sum, Figure 10), not the joint source entropy itself. (F's larger MAE of 1.115 is off this coordinate scale — the corpus uses four feed values, `F7.3`, `F13.3`, `F22`, and `F40` inch/min ($\approx$185–1016 mm/min) — and is not a recovery failure: its integer digits are recovered, with the per-axis bit budget crediting F 4.5/4.9 bits.) The defensible characterization is therefore a *coarse-resolution coordinate recoverer* bounded at the uniform-noise ceiling on the hundredths–thousandths, where CAM interpolation residuals dominate.

The teacher-forced upper-bound caveat.

The entropy–accuracy account above is computed under teacher-forced (TF) evaluation, where each recovery step is conditioned on the true prefix. TF is the correct comparison to a source-entropy ordering — it isolates per-position recoverability from autoregressive error propagation — but it is an *upper bound* on deployment recovery. Under autoregressive (AR) inference the companion decoder's digit head collapses to $\approx$0.10 accuracy by exposure-bias mode-collapse [1]. The ordering characterized here is therefore the in-principle ceiling under ideal conditioning; AR deployment recovery is strictly below it, and any deployment claim must read the TF numbers as the optimistic boundary, not the operating point.

8. The Numeric Target-Encoding Artifact: Corrected, Retrained, and Re-Derived

An earlier draft of this work reported the fine-digit numbers on a checkpoint whose numeric target encoding contained a float-epsilon artifact, disclosed at the time as a pending pre-patch caveat. That caveat is now *resolved*: we corrected the encoding, retrained the model 5-fold on the corrected targets, and re-derived the entropy and accuracy under a single consistent round-encoding. We report the resolution here because it converts the deepest objection — that the fine-digit story rested on a known encoding bug — into a strengthened result.

The artifact.

The historical `encode_values_to_digits` routine iteratively multiplied the decimal part by ten and truncated (`slot[s] = int(dec_part * 10) % 10`), which is exact in IEEE 754 only for exactly representable values. For example the authored facing coordinate `3.2750` is stored as 3.27499 . . ., so the iterative chain emits a spurious 9 at the ten-thousandths and a 4 rather than 5 at the thousandths, giving digits $[0, 3, 2, 7, 4, 9]$ instead of the authored $[0, 3, 2, 7, 5, 0]$. The net effect was to *mislocate the precision floor by one decimal place*: it manufactured a spurious digit-9 mass at the ten-thousandths (making slot-5 look high-entropy and "three-regime"), while corrupting the genuine thousandths digit. The corpus-weighted impact is large: re-encoding every numeric position with the corrected routine changes at least one digit slot on 32.5% of numeric positions (25,765 of 79,345), of which 92.8% (23,909) are exactly the terminal-zero $9 \to 0$ fixes at the finest slot (`audit/patched_numeric_analysis.json`, key `corpus_target_change_rate`). The target *tokens* are otherwise identical between the released and patched runs, so this is a clean controlled comparison.

The fix, retrain, and re-derivation.

A round-to-scaled-integer rewrite (scaled $=$ round$(|v| \cdot 10^4)$ once, then floor-divide per slot) is committed to `src/miracle/model/digit_value_head.py`; the verification harness (`scripts/analysis/verify_digit_encoding_fix.py`) confirms zero corruptions on the same positions. We then retrained the full system 5-fold on the corrected targets (`checkpoints/full_window_5fold_patched/`, released untouched alongside the original). **Every quantitative fine-digit result in this paper is computed on the patched-encoder retrain.** The headline categorical accuracies are robust to the patch: token 0.774 vs. 0.781, numeric 0.570 vs. 0.585, command 0.891 vs. 0.888, type 0.983 vs. 0.984, parameter-type 0.993 vs. 0.993 (patched vs. released, all drift $\leq$1.5 pp). The fine-digit *structure*, by contrast, moves exactly as the bug predicts: the spurious slot-5 three-regime collapses, and the true precision floor reappears at the thousandths (Sections 4–5). On the digit-value head the corrected ten-thousandths is the authored terminal digit; on the V8 grammar tokenizer the X/Y/Z ten-thousandths is discarded by the 0.001 precision grid, so its recovery accuracy is 0.97–1.00 for every op-class (it is a tokenizer constant, Table 1), not the spurious high-entropy "three-regime" band of the buggy encoding.

Why the correction strengthens the result.

The decoder-free source-entropy apparatus of Section 4 — which never touches the model encoding — independently locates the ceiling at the hundredths–thousandths, the coordinate-repeat structure on facing X (2.10 bits), the cutting-depth quantization on Z, and the arc-radius vocabulary on R, and cleanly separates the near-uniform source ten-thousandths from its constant tokenizer representation (Table 1). With the model target now encoded consistently with that source, the lead entropy–accuracy relationship is not merely preserved but *stronger*: the per-(axis, slot) Pearson r improves from -0.68 (with its known cross-axis encoding inconsistency) to -0.83 (Spearman $\rho = -0.91$, $n = 30$), and the pooled six-slot relationship to $r = -0.89$ ($\rho = -0.94$). The prior slot-5 "three-regime" and the "96.5% of face X from four overruns at slot-5" were encoding artifacts and have been removed; the genuine three-regime structure survives, correctly located at the thousandths. The float-epsilon issue is therefore no longer a caveat on the result — it is the mechanism whose correction sharpened it.

9. Vocabulary Cardinality and the Entropy Floor

The numeric vocabulary cardinality K controls how finely the continuous coordinate source is quantized into discrete tokens, and is therefore directly relevant to where the fine-digit source entropy sits in the recoverability ordering. We characterize structural recovery across $K \in \{24, 69, 335, 2{,}418\}$ plus a methodology-matched $K = 2{,}418$ control (T3.1); Table 7 and Figure 12 report the five-point spectrum.

Provenance and division of labor with Paper A.

The vocabulary-cardinality variants are decoder-architecture experiments shared with the companion decoder paper [1]; we reproduce only the single K-spectrum panel that bears directly on the entropy floor and cross-cite [1] for the full ablation, the per-fold detail, and the schedule-confound analysis, rather than re-plotting those shared assets here.

The sweet-spot claim is withdrawn.

An earlier presentation reported a $K = 335$ "sweet spot" with a $+11.1$ pp AR-token advantage over the headline. That comparison was a confound: the smaller-K variants were trained under a different schedule (scheduled-sampling SS=0, digit-loss weight dw=0) than the headline (SS=0.5, dw=1.0). The matched-methodology $K = 2{,}418$ control (T3.1, same SS=0/dw=0) is the strict maximum on every metric (AR struct token 0.470 ± 0.034 vs. headline 0.317 ± 0.116, $+15.2$ pp, 5/5 folds positive), and within the matched family $K = 335$ falls -4.1 pp *behind* the matched $K = 2{,}418$ (paired $t(4) = -0.85$, $p = 0.44$, n.s.). The earlier advantage was the schedule, not the cardinality; we retract the sweet-spot framing.

Within-methodology cardinality effect.

Across the four matched variants, AR struct token is monotonically non-decreasing in K $(0.405, 0.415, 0.429, 0.470)$; the omnibus is significant on the exact (2×10^5-permutation Monte-Carlo) Friedman test ($p = 0.0018$ for AR token, $p = 0.019$ for AR sequence; `audit/k_spectrum_stats.json`, key `rm_anova`), reported in preference to the asymptotic χ^2 ($p = 0.0076$ / 0.022), which is unreliable at $n = 5$ blocks. Larger K is mildly better within the matched family, but the effect is small relative to the $+15$ pp schedule effect. The interpretation for the recoverability ordering is that finer numeric quantization does not lower the source entropy on the fine decimals — the thousandths digit is near-uniform regardless of how the vocabulary buckets it — it only changes the placeholder granularity of the structural stream.

10. Audit-Leverage Corollary

The source-entropy ordering carries a direct corollary for sensor-side forensic audit of a CNC run, complementing cryptographic file-hash signing and controller attestation along an entropy axis. Operation classes with low coordinate entropy — facing-pass workpiece-edge repeats, and the shallow Z rapid-vs-cut depth structure — concentrate most of their coordinate mass on a few specific values; any deviation from that modal pattern is statistically conspicuous and ideal for sensor-side audit. Quantitatively, facing concentrates $\approx 63\%$ of its X-coordinate mass in four values (`X-0.1254` 27%, `X3.5354` 21%; `audit/raw_gcode_source_entropy.json`), against near-uniform thousandths for adaptive HSM, and a class-conditional modal-command predictor reaches 0.811 (`audit/class_conditional_modal.json`) — the measurable gradient that makes low-entropy classes auditable. This remains a corollary of the ordering, not a validated detector. Operation classes with high coordinate entropy — adaptive HSM and tight pocketing, where the postprocessor

Table 7. Structural recovery across the numeric-vocabulary spectrum $K \in \{24, 69, 335, 2{,}418\}$ plus the T3.1 methodology-matched $K = 2{,}418$ control. *Structural* restricts the metric to command-code and axis-letter positions (numeric / placeholder positions excluded). 5-fold mean ± std, full-window targets, under teacher-forced (TF) and autoregressive (AR) evaluation; AR-token bootstrap 95% percentile CIs from 10^4 resamples in brackets. The T3.1 control — $K = 2{,}418$ trained under matched SS=0/dw=0 — *achieves the highest AR structural performance of the five measured variants* (AR token 0.470±0.034; +15.2 pp paired across folds vs. the headline, 5/5 folds positive; AR sequence 0.128±0.019, +7.1 pp, 5/5 folds, paired $t(4) = 7.70$, Holm-adjusted $p = 0.006$, Hedges' $g_z = 2.76$). Within the matched-methodology family {T3.1, $K = 335$, $K = 69$, $K = 24$}, K=335 falls −4.1 pp behind the matched $K = 2{,}418$ on AR token (paired $t(4) = -0.85$, $p = 0.44$, n.s.). The +11.1 pp K=335 advantage over the original headline is therefore the SS/dw training-schedule change, not vocabulary cardinality.

Variant	K	TF token	AR token [95% CI]	TF seq exact	AR seq exact
$K = 2{,}418$ (4-digit) headline†	2,418	0.960 ± 0.020	0.317 ± 0.116 [0.22, 0.40]	0.213 ± 0.048	0.058 ± 0.039
$K = 2{,}418$ (4-digit) **T3.1 matched control**‡	2,418	$\mathbf{0.978 \pm 0.015}$	$\mathbf{0.470 \pm 0.034}$ [0.44, 0.50]	$\mathbf{0.344 \pm 0.040}$	$\mathbf{0.128 \pm 0.019}$
$K = 335$ (2-digit)	335	0.977 ± 0.014	0.429 ± 0.123 [0.33, 0.51]	0.287 ± 0.050	0.081 ± 0.045
$K = 69$ (1-digit)	69	0.965 ± 0.019	0.415 ± 0.036 [0.39, 0.44]	0.250 ± 0.031	0.051 ± 0.034
$K = 24$ (placeholder, Design B)	24	0.927 ± 0.024	0.405 ± 0.143 [0.28, 0.50]	0.068 ± 0.017	0.066 ± 0.014

†Headline: scheduled-sampling SS=0.5, digit-head loss weight dw=1.0. ‡T3.1 control: identical $K = 2{,}418$ vocabulary, matched SS=0/dw=0 training schedule. The remaining rows ($K \in \{24, 69, 335\}$) also use SS=0/dw=0, so the four non-headline rows form a clean four-point vocabulary-cardinality contrast under one training methodology.

emits effectively-random thousandths digits — are unauditable by sensor-side recovery alone: a maliciously substituted coordinate is indistinguishable from a legitimate one because the legitimate distribution itself approaches the uniform-noise ceiling (Section 5). Sensor audit and cryptographic verification are therefore complementary along the operation-class entropy axis: sensor audit is most useful where humans are most blind (predictable repetitive operations), and cryptographic hashing remains the only viable defense where sensor audit is information-theoretically blind (high-entropy optimizer-generated paths). The same ordering holds per axis: X-coordinate audit is most informative on facing operations, Z-coordinate audit on its shallow rapid-vs-cut depth structure. This is a corollary of the ordering, scoped to this corpus, not a validated detector; the companion decoder paper [1] reports the corresponding tamper-detection operating points and their open-vocabulary collapse.

11. Limitations

Encoding patch (resolved) and source-vs-tokenized reconciliation.

The float-epsilon target-encoding artifact disclosed in earlier drafts is resolved: the encoding was corrected, the model retrained 5-fold, and all fine-digit figures re-derived under a consistent encoding (Section 8). The headline categorical accuracies drift ≤ 1.5 pp from the released checkpoint, and the lead 30-cell relationship is *stronger* on the corrected data ($r = -0.83$). We also reconciled the earlier "three-decimal corpus / structural-zero ten-thousandths" framing against the source: the corpus is four-decimal, and the X/Y/Z ten-thousandths zero is a tokenizer representation artifact, not a source or sensor limit (Section 3, Table 1). The source-apparatus and patched figures (the source-vs-tokenized table, the per-(axis, slot) and op-class entropy heatmaps, the per-axis digit-value histograms, the bit-budget and bathtub panels, and the $n = 30/n = 54$ scatters) are regenerated from the authoritative audit JSONs (`figures/regenerate_source_apparatus_figures.py`, `figures/regenerate_patched_figures.py`); the remaining displayed legacy panels (confusion matrices, learning curves) are unaffected by the patch.

Single corpus, single platform, single material.

All results are conditional on one corpus from a single desktop CNC platform, a single workpiece-material class, and $n = 5$ executions partitioned by file-level cross-validation. Whether the three-regime decomposition is platform-specific or a general property of CAM-generated toolpaths is the principal open question. A second corpus — a second machine, or even the same parts re-exported through a different CAM postprocessor (entropy-only, no machining) — would test generality; until it lands, the ordering is scoped to this corpus and not claimed as general.

Closed-vocabulary, in-distribution.

The entropy ceiling is computed at the empirical training distribution and the recovery confirmation is closed-vocabulary (97.6% of test distinct G-code lines, and 99.0% of test line-token occurrences, appear in the training split; per-row means, `audit/train_test_gcode_overlap.json`). Open-vocabulary or novel-operation-class deployment may exhibit different per-head entropy profiles; the leave-one-class-out floor reported by the companion decoder [1] bounds this risk on the present corpus.

Field coverage.

Spindle speed is single-valued (`S12000`) and contributes 0 bits (a deterministic field, not absent); a single cutting tool (`T2`) is used throughout, so tool number is likewise deterministic (0 bits); the corpus uses R-notation arcs only (no I/J interpolation) and has thin feed-rate (F) support ($n = 16$ raw over the nine authored toolpaths / 159 model-side over the windowed 5-fold test split). Entropy figures for I, J, and K (captured by the extraction regex but with no support in this corpus) are not estimable (genuinely absent); F figures are reported at thin support and are noisier than the X/Y/Z core.

Marginal entropy, not a conditional bound.

The reported relationship is an ordering keyed to the marginal $H(X)$, not a tight Fano bound (Section 2.1); only the 0-bit slots and the 4 Hz sub-Nyquist floor are genuine limits. A conditional-entropy or channel-MI estimate is a natural follow-on.

Teacher-forced upper bound.

The entropy–accuracy relationship is a TF upper bound (Section 7); AR deployment recovery on the numeric head is at floor (≈ 0.10). The ordering describes the in-principle ceiling under ideal conditioning, not the deployment operating point.

12. Conclusion

We characterized a source-entropy-correlated recoverability ordering for low-rate sensor reconstruction of CNC toolpath coordinates, anchored by two genuine limits: the 0-bit source slots, and a 4 Hz sub-Nyquist physical floor that aliases 77.7% of G-code blocks. The Shannon source entropy of the CAM-authored four-decimal coordinate corpus has a sharp, physically interpretable structure — integer places near-deterministic, fine decimals plateauing near the $\log_2 10 \approx 3.32$-bit uniform-noise ceiling across the hundredths–thousandths (10^{-2}–10^{-3}, the genuine source precision floor). The X/Y/Z ten-thousandths is equally uniform in the source; its appearance as a "structural zero" is a tokenizer representation artifact, not a source or sensor limit (Table 1). The fine-digit ceiling stratifies into CAM-driven source regimes at the thousandths (facing 2.10 bits, pocketing 3.32 bits $\approx$ adaptive HSM 3.31 bits), established decoder-free on raw G-code source. We measure the marginal entropy $H(X)$, not a conditional Fano bound, and report the empirical ordering: on a float-epsilon-corrected, 5-fold-retrained checkpoint, recovery accuracy is inversely related to source entropy over 30 per-(axis, slot) cells under a consistent encoding ($r = -0.83$, Spearman $\rho = -0.91$; pooled six-slot $r = -0.89$, illustrative at $n = 6$). The ordering is cross-head (the categorical command head is near-ceiling and flat);

two independent recoverers — a companion decoder [1] and a from-scratch baseline — land at the same ceiling on the categorical command head (0.877 $\approx$ 0.888), but not on the numeric head, where the baseline exceeds the decoder by 18.4 pp under teacher forcing, so we make no architecture-invariance claim there. The encoding artifact disclosed in earlier drafts is resolved by the retrain — the prior slot-5 three-regime structure was an artifact, now removed, and the corrected relationship is stronger — and the headline categorical accuracies are robust to the patch ($<$ 1.5 pp drift). We report the single-corpus / single-platform / closed-vocabulary scope and that the relationship is a teacher-forced upper bound. The registered next steps are a conditional-entropy / channel-MI estimate to convert the ordering toward a tight bound, and a second corpus to test whether the regime structure is general or corpus-specific.

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Data Availability Statement: The source-entropy apparatus (`scripts/analysis/raw_gcode_source_entropy.py`), the patched-encoder figure regeneration script (`figures/regenerate_patched_figures.py`), and the audit JSONs that ground every reported number (`audit/patched_numeric_analysis.json` (authoritative for all fine-digit numbers), `audit/raw_gcode_source_entropy.json`, `audit/per_head_per_opclass_entropy_5fold.json`, `audit/gcode_row_durations.json`) will be deposited in a public Zenodo archive with the companion artifact at proof stage, together with the patched checkpoints (`checkpoints/full_window_5fold_patched/`); the Zenodo DOI will be inserted at the proof stage.

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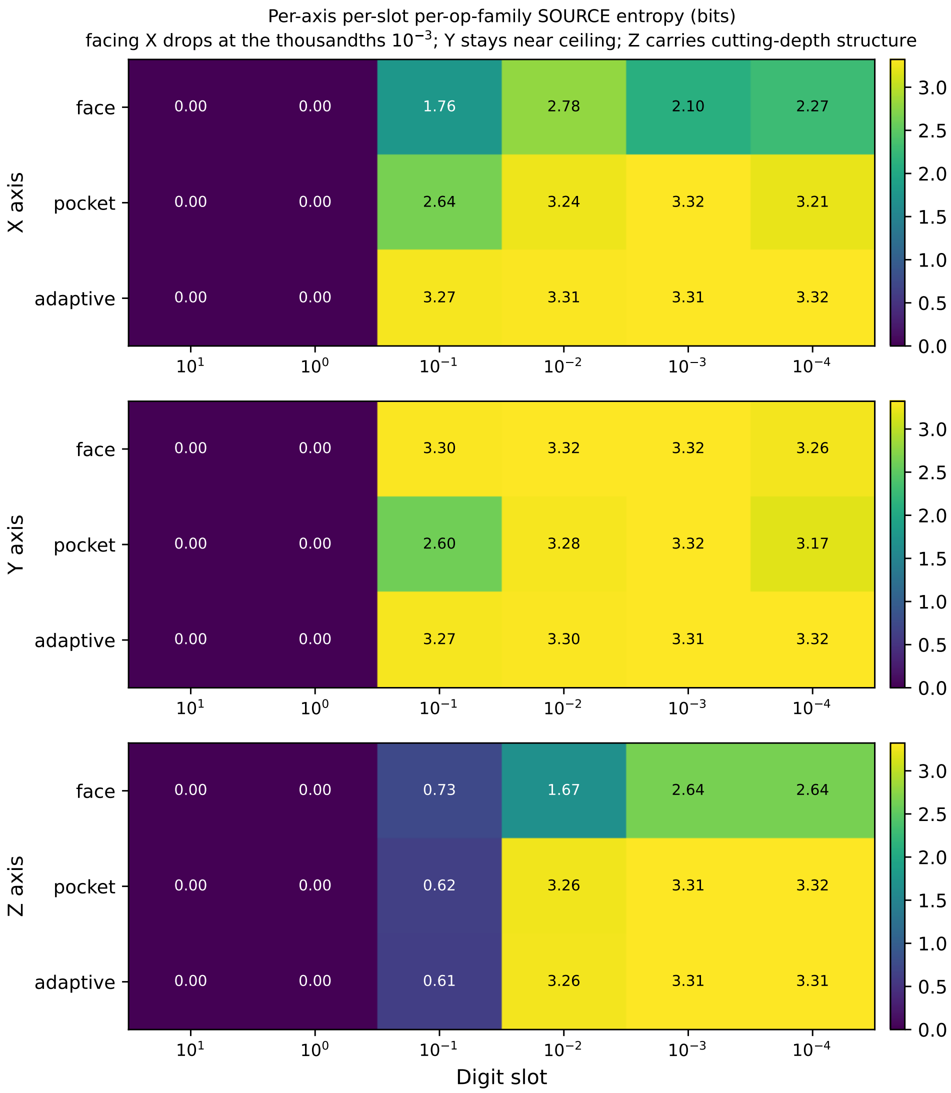
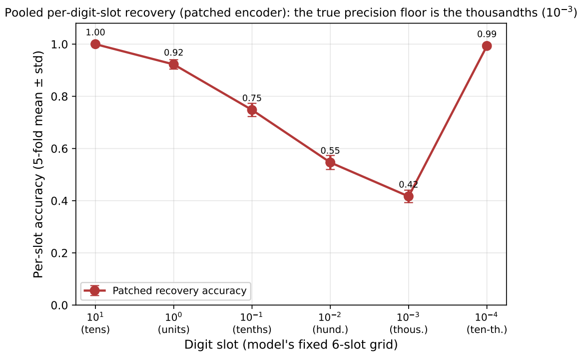
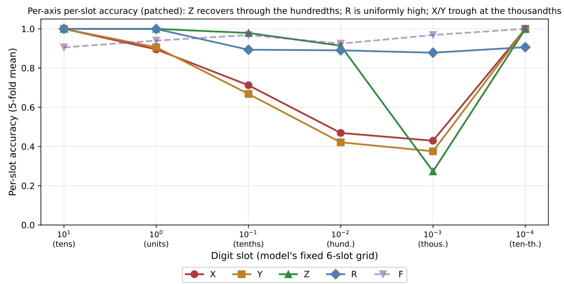


Figure 6. Per-axis (X, Y, Z top to bottom) per-digit-slot per-op-family *source* entropy. The X–Y–Z asymmetry is sharpest at the *thousandths* slot (10^{-3}): facing X drops to $\approx$ 2.1 bits (workpiece-edge repeats), while Y stays near the ceiling; Z carries concentrated cutting-depth structure at the tenths (0.6 bits) and approaches the ceiling by the hundredths. Values are SOURCE entropy on the four-decimal corpus; the near-uniform X/Y/Z ten-thousandths shown here is discarded by the 0.001 tokenizer grid (Table 1). Regenerated from `audit/raw_gcode_source_entropy.json` (`figures/regenerate_source_apparatus_figures.py`).

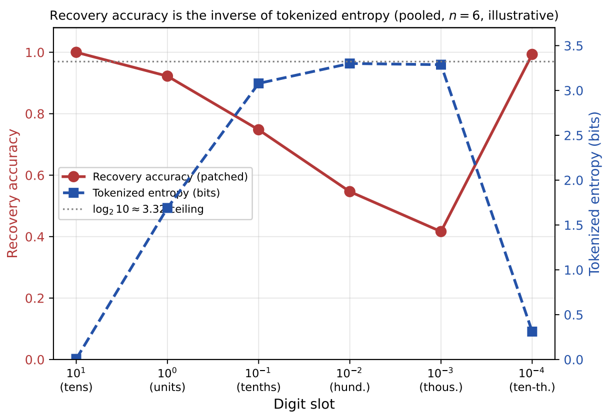


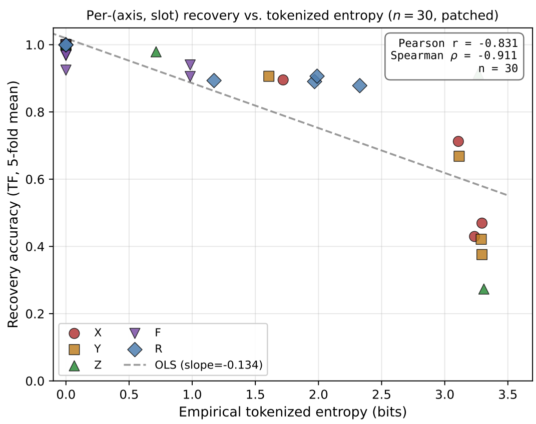


(b) Per-axis overlay (X/Y/Z/R/F).

(a) Pooled per-digit-position accuracy.

Figure 7. The numeric-recovery "bathtub" (patched encoder). (a) Pooled per-digit-position accuracy (5-fold mean $\pm$ std, teacher-forced): slot 0 at 1.00, descending monotonically to the *thousandths* trough at 0.42 (the true precision floor), then slot 5 (ten-thousandths) recovering to 0.99 as a tokenizer constant (a representation artifact; Table 1), not a fine-digit recovery. (b) Decomposed per axis: X/Y descend to a thousandths trough, Z recovers through the hundredths (cutting-depth quantization), R is uniformly high (arc-radius vocabulary). Regenerated from `audit/patched_numeric_analysis.json`.





(a) Pooled per-slot overlay (illustrative, $n = 6$). **(b)** Per-(axis, slot) scatter (lead, $n = 30$).

Figure 8. Tokenizer-representable entropy orders the numeric-recovery profile (patched encoder, encoding-consistent; both axes are the tokenized object the model can represent, see Section 7). (a) Per-slot accuracy (red) overlaid on per-slot tokenized entropy (blue); recovery is the inverse of the entropy profile, with the dashed line at the $\log_2 10 \approx 3.32$-bit ceiling. Reported as illustrative only ($n = 6$; pooled $r = -0.89$, $\rho = -0.94$; bivariate normality uncheckable). (b) Per-(axis, slot) cell-level scatter (30 cells, marker by axis); the cell-level Pearson $r = -0.83$ (Spearman $\rho = -0.91$) is the load-bearing statistic. Both regenerated from `audit/patched_numeric_analysis.json`.

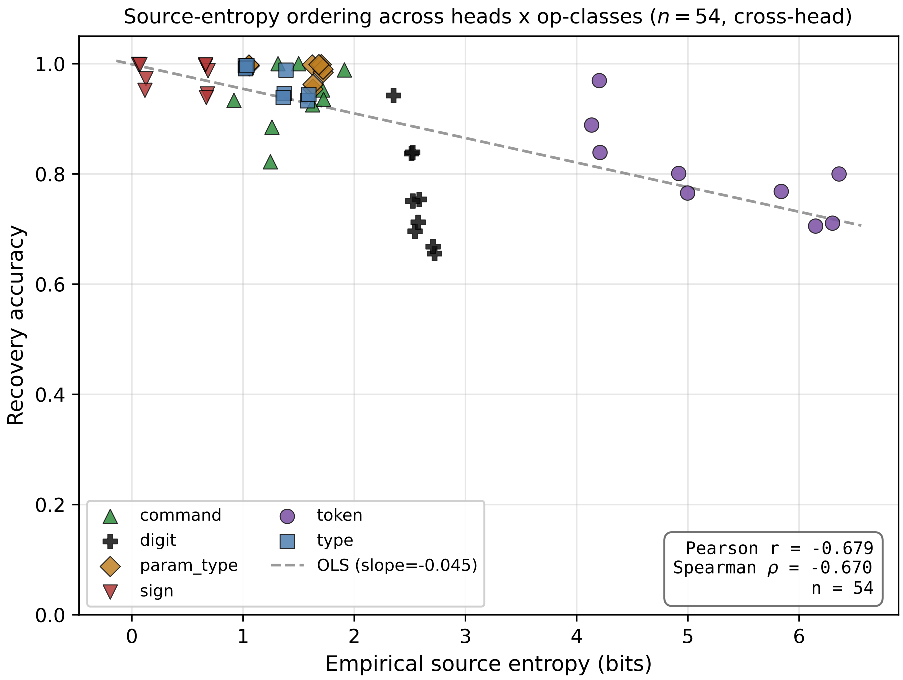


Figure 9. The source-entropy ordering across heads and operation classes: 54 per-(head, op-class) cells, measured recovery accuracy against empirical source entropy (Pearson $r = -0.68$, Spearman $\rho = -0.67$, $n = 54$). The negative slope is *cross-head* — it is the contrast between near-ceiling categorical heads (command, parameter-type, sign, individually flat) and the high-entropy numeric heads (digit, $r \approx -0.9$) that produces it, not a within-head bound. This panel uses live per-cell accuracies on the released checkpoint ($r = -0.68$, which supersedes the $r = -0.55$ obtained before the head$\times$op-class alignment repair); the load-bearing, encoding-consistent statistic is the $n = 30$ per-(axis, slot) scatter (Figure 8b, $r = -0.83$). Regenerated from `audit/entropy_accuracy_correlation_fixed.json` after the head$\times$op-class alignment repair (`figures/regenerate_source_apparatus_figures.py`).

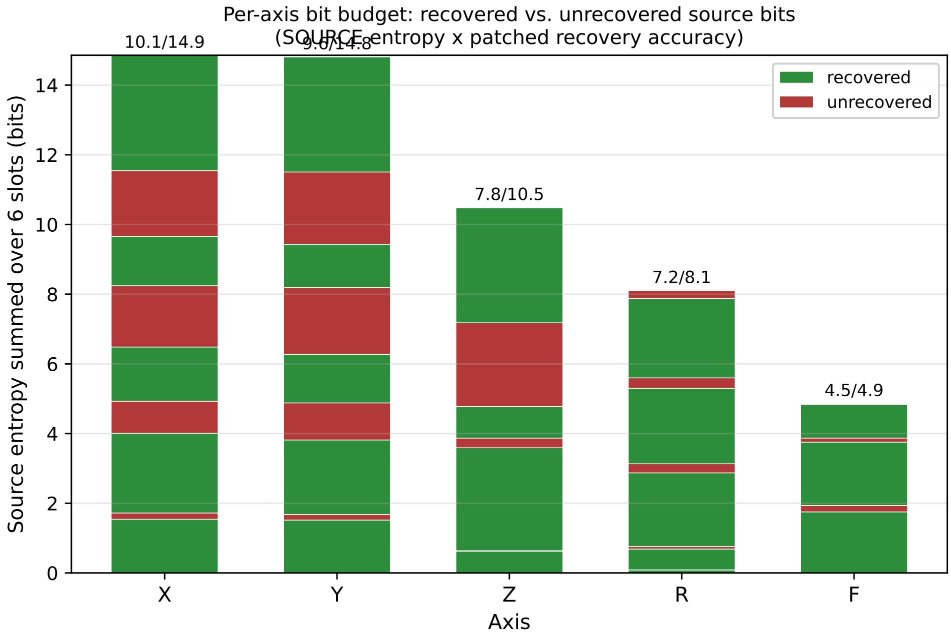
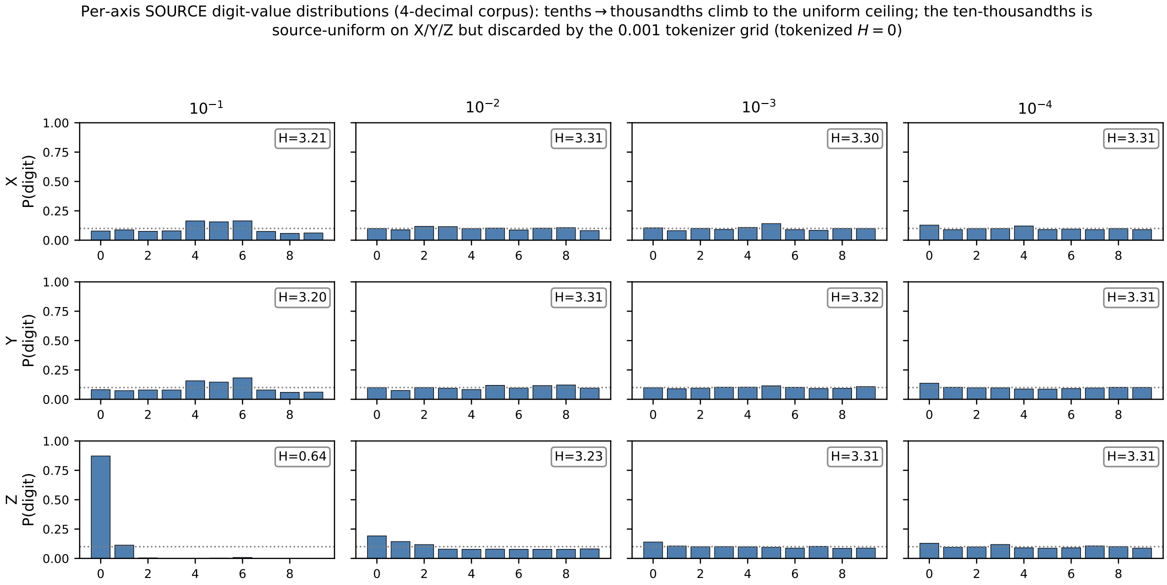
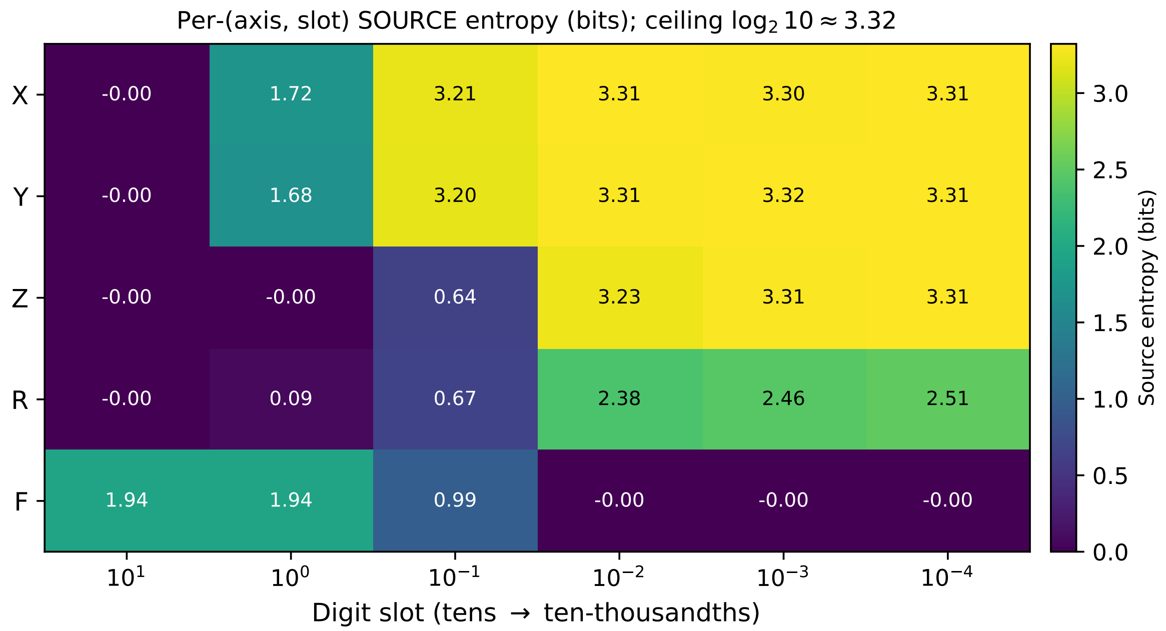


Figure 10. Per-axis "bit budget": total *source* entropy summed across the six slots (bar height), split per slot; green/red shading approximates recovered ($\approx H_s\, a_s$) vs. unrecovered ($\approx H_s(1 - a_s)$) bits for SOURCE slot entropy H_s (`audit/source_vs_tokenized_entropy.json`) and patched 5-fold accuracy a_s. R (7.2/8.1) recovers nearly all of its small discrete-vocabulary budget and F (4.5/4.9) its integer budget; X (10.1/14.9) and Y (9.6/14.8) lose the bulk of their unrecovered bits to the uniform hundredths–thousandths digits; Z (7.8/10.5) recovers mostly the cutting-depth structure at slots 0–3. Regenerated from `audit/source_vs_tokenized_entropy.json` and `audit/patched_numeric_analysis.json` (`figures/regenerate_source_apparatus_figures.py`).



(a) Per-axis (X/Y/Z) per-slot digit-value distributions.



(b) Per-(axis, slot) entropy and modal-fraction map (all five fields).

Figure 11. Source digit-value structure (four-decimal corpus). (a) Empirical SOURCE digit distributions per axis (X/Y/Z) $\times$ slot (10^{-1} to 10^{-4}), Shannon entropy annotated; the tenths/hundredths/thousandths climb to near-uniform and *plateau* from the hundredths (10^{-2}) onward at the $\approx$ 3.3-bit ceiling (the CAM-interpolation signature and true precision floor). The ten-thousandths (10^{-4}) is equally uniform in the source ($\approx$ 3.31 bits), becoming a structural zero only under the 0.001 tokenizer grid (a representation artifact; Table 1); Z's 10^{-1} is the cutting-depth-step exception. (b) Per-(axis, slot) SOURCE entropy heatmap (X/Y/Z/R/F): dark = recoverable/low-entropy, bright = uniform-noise ceiling. Both regenerated from `audit/raw_gcode_source_entropy.json` and `audit/source_vs_tokenized_entropy.json` (`figures/regenerate_source_apparatus_figures.py`); the corrected per-(axis, slot) entropies are also tabulated in Tables 1–2.

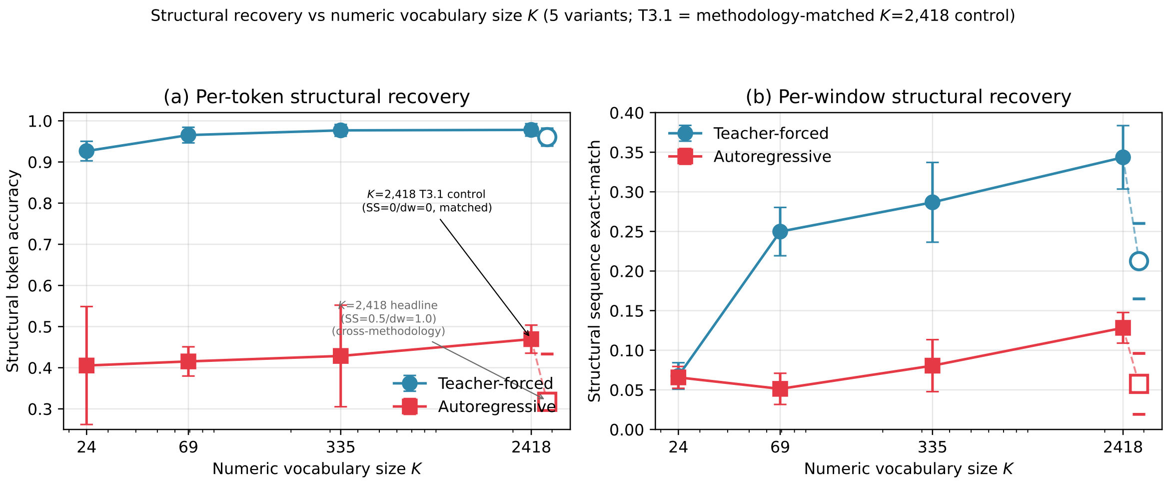


Figure 12. Structural recovery vs. numeric vocabulary size K across five variants. Filled markers / solid line: matched-methodology $K \in \{24, 69, 335, 2{,}418\}$ (SS=0, dw=0). Open marker: $K = 2{,}418$ headline (SS=0.5, dw=1.0). (a) AR per-token: matched $K = 2{,}418$ is the maximum; the $K = 335$ advantage over the headline reflects the schedule, not cardinality. (b) AR per-window exact-match: matched $K = 2{,}418$ dominates (Holm $p = 0.006$).